

GIFT University Gujranwala

Project Final Report

For

**Smart EV Charging and Navigation Management
System**

Submitted By:

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1 Introduction

The proposed project focuses on developing a mobile and backend system that serves as a centralized platform for electric vehicle (EV) users in Pakistan. It aims to simplify EV usage by helping users find nearby charging stations, plan optimized routes, and manage their vehicles efficiently. Currently, in Pakistan, the adoption of electric vehicles is increasing, but the supporting digital infrastructure for locating charging stations and managing EV usage remains limited. Most existing solutions either focus on international networks or lack localized data, making it difficult for users to find reliable and affordable charging options. This project aims to address these challenges by providing accurate information about nearby charging stations, real-time cost and availability updates, and intelligent route planning based on vehicle battery levels and range. The gap being filled is the absence of a dedicated, user-friendly mobile app tailored to the needs of Pakistani EV users. The objectives include integrating local station data with map-based navigation, allowing users to manage their vehicles, and enabling secure profile and authentication features. Furthermore, the app seeks to enhance convenience through optimized trip planning and promote eco-friendly transportation across Pakistan. Overall, this project will serve as a significant step toward supporting the country's transition to sustainable mobility by simplifying EV usage for local drivers.

1.1 Project Background

- The project aims to create a digital platform for electric vehicle (EV) users in Pakistan. It will help users locate nearby charging stations and plan efficient travel routes. As global EV adoption increases, many countries already use integrated mobile apps connected to charging networks. In Pakistan, the charging infrastructure is scattered and lacks visibility for users. EV owners face challenges such as difficulty finding stations, comparing costs, and planning trips. The proposed solution includes a mobile app and backend system that centralizes EV-related data.
- Key features will include station location mapping, cost comparison, route optimization, and vehicle management.
- The project's goal is to make EV usage in Pakistan more convenient, reliable, and practical.

1.2 Project Scope

The scope of this project is to design and develop a mobile and backend system that serves as a centralized platform for electric vehicle (EV) users in Pakistan to locate charging stations, plan efficient routes, and manage vehicle-related data. The proposed system will allow users to search for nearby charging stations based on their current location, view station details such as availability, type of charger, and cost per unit, and compare these options to make informed decisions. It will also enable users to plan optimized routes by considering battery range, distance,

and station locations along the journey. The system will include a user-friendly mobile application that communicates with a secure backend database, ensuring accurate and up-to-date station information. Administrators will have access to a dedicated portal to manage station data, update costs, and monitor network activity to maintain data reliability. The project will support features such as user registration, login authentication, and profile management for personalized experiences. Additionally, users will be able to create and manage multiple vehicle profiles.

The project will primarily focus on data accuracy, route optimization, and user convenience, ensuring seamless interaction between users and the system. The system will use GPS integration and APIs to fetch real-time data, enhancing precision in station location and route planning. However, the scope does not include the development or installation of physical charging stations or hardware components; it strictly focuses on software-based solutions. It will also exclude payment gateway integration in its initial phase, though it can be added in future upgrades. Furthermore, the system will not handle third-party charging network management directly but will provide data synchronization options for registered stations. In summary, the project aims to deliver a dependable, localized digital platform that simplifies EV charging in Pakistan, encourages sustainable transportation, and enhances user trust in EV technology.

1.3 QA Testing Overview

This portfolio presents the **Software Quality Assurance (SQA) and testing activities** performed for the **Smart EV Charging and Navigation Management System**. The testing was conducted to verify that the system functions according to its specified requirements and that individual modules work correctly both independently and when integrated with other modules. The testing activities covered **manual testing, functional testing, unit testing, integration testing, system testing, acceptance testing, usability testing, and requirement traceability**. These activities were performed to evaluate the functionality, reliability, usability, and overall quality of the system.

2 Testing and Evaluation

This chapter may include the following sections. (Students are required to perform the testing both manually and automatically).

2.1 Test Plan

This Test Plan defines the testing strategy, scope, objectives, and responsibilities for the Smart EV Charging and Navigation Management System. The system is a Flutter-based mobile application supported by a Firebase backend and a FastAPI-served DistilBERT machine learning module. The purpose of this plan is to ensure that all functional requirements are implemented correctly, all modules integrate without errors, and the final product meets user expectations.

This plan covers all ten functional modules of the system: User Authentication, Charging Station Locator, Route Planning, Vehicle Management, Cost Comparison and Estimation, Station Owner Portal, EV Owner Portal, Admin Management Portal, Rating and Review with ML Sentiment Analysis, and Amenities. Testing will be carried out through manual unit testing, integration testing, system testing, acceptance testing, and usability testing.

2.2 Product Scope

The Smart EV Charging and Navigation Management System is a mobile application designed for electric vehicle owners, charging station owners, and system administrators in Pakistan. The application addresses the lack of a centralized, intelligent platform for locating EV charging stations and planning energy-efficient routes.

The system provides the following core functionalities:

- EV owners can register, log in, search for nearby charging stations on an interactive map, filter by charger type (AC/DC), price, and area, and get directions.
- EV owners can plan routes using A* and Yen's K-Shortest Paths algorithms with OSRM real-road data, and receive automatic charging stop recommendations based on battery level.
- EV owners can add and manage vehicle profiles and compare estimated charging costs across different stations.
- EV owners can submit written reviews which are analyzed using a fine-tuned DistilBERT sentiment model (Positive / Neutral / Negative) served through a FastAPI backend.
- Station owners can register and manage charging stations, update availability status, manage charger units, and view usage reports and session history.
- Administrators can manage all user accounts, approve or reject station registrations, monitor and remove inappropriate reviews flagged by the ML model, and maintain audit logs.

The system does not support real-time slot booking, payment processing, or hardware-level charger integration in this version. These are identified as future work items.

2.3 Intended Audience

This Test Plan is intended for the following stakeholders: the project supervisor and academic panel at GIFT University who will evaluate the system during the final presentation and viva; the development and testing team consisting of the four group members who are responsible for executing test cases and reporting results; and the project committee members who will review this document as part of the Capstone 2 deliverable. The document assumes that readers are familiar with basic software testing terminology and the functional requirements of the Smart EV Charging and Navigation Management System as specified in the SRS document.

2.4 Definitions, Acronyms and Abbreviations

The following terms are used throughout this Test Plan:

Term / Acronym	Definition
AC	Alternating Current one of the two charger types supported by the system
DC	Direct Current fast charging type supported by the system
API	Application Programming Interface used for OSRM routing and FastAPI ML backend
EV	Electric Vehicle
FastAPI	Python web framework used to serve the DistilBERT ML model
Firebase	Google's cloud platform used for authentication, Firestore database, and real-time sync
FR	Functional Requirement defined in the SRS document
GPS	Global Positioning System used for live user location tracking
ML	Machine Learning used for sentiment analysis in the Review module
OSRM	Open Source Routing Machine free routing API used for real road route planning
TC	Test Case an individual test scenario with inputs and expected results
UC	Use Case a functional scenario defined in the SRS
UI	User Interface the screens and interactions of the Flutter app
STLC	Software Testing Life Cycle
UAT	User Acceptance Testing
Pass	Test result indicating the system behaved as expected
Fail	Test result indicating the system did not behave as expected

2.5 Test Pass/ Fail Criteria

The following criteria are used to determine whether a test case has passed or failed:

Pass Criteria

- The system produces the expected output as defined in the test case for a given input and precondition.
- All High Priority test cases are completed and pass without any critical defects.
- All functional requirements (FR-1 to FR-51) are covered by at least one passing test case.
- The DistilBERT sentiment API returns the correct sentiment class (Positive / Neutral / Negative) for sample review texts with at least 90% accuracy on test inputs.
- Route planning correctly inserts charging stops when the destination exceeds the vehicle's remaining battery range.

- Admin, Station Owner, and EV Owner portals are accessible only to their respective roles no unauthorized access is possible.

Fail Criteria

- The system produces an incorrect, unexpected, or no output for a valid input.
- Any High Priority test case fails the defect must be logged and fixed before release.
- Any security-related test fails for example, a user of one role accessing another role's screens.
- The ML API returns an incorrect sentiment label for clearly positive or clearly negative sample reviews.
- The application crashes, freezes, or throws unhandled exceptions during any test case execution.

A defect is logged for every failed test case. High and Medium priority defects must be resolved before final submission. Low priority defects are documented as known issues.

2.6 Verification

Verification ensures that the Smart EV Charging and Navigation Management System has been built correctly according to the specified requirements and design documents. For this project, verification activities include:

- Reviewing the Flutter source code to confirm that each functional requirement (FR-1 to FR-51) is implemented in the corresponding module.
- Confirming that the route optimization service implements A* (shortest and fastest variants) and Yen's K-Shortest Paths (k=4) as specified in the algorithm section of the SRS.
- Verifying that the FastAPI backend correctly loads the fine-tuned DistilBERT model from the saved .safetensors checkpoint and returns sentiment predictions with confidence scores.
- Confirming that Firebase Firestore correctly stores review text, star rating, sentiment label, confidence score, flag status, and admin action status as specified.
- Checking that role-based access control is enforced EV Owner, Station Owner, and Admin each see only their respective screens.
- Comparing the implemented use case flows with the sequence diagrams (SED-EV01 to SED-EV17) defined in the SRS to confirm design-to-code consistency.

2.7 Validation

Validation ensures that the system actually meets the real-world needs of its intended users — EV owners, station owners, and administrators. Validation for this project is carried out through:

- End-to-end scenario testing simulating a complete user journey from registration through station search, route planning, navigation, and review submission to confirm the workflow is smooth and logical.
- Acceptance testing with realistic test data using actual GPS coordinates, real vehicle battery ranges (e.g. 300 km range, 60 kWh battery), and EV-specific review texts to validate that the system produces meaningful outputs.
- Sentiment validation submitting sample reviews such as 'The DC fast charger worked perfectly and the station was clean' (expected: Positive) and 'The charger was broken and nobody was there to help' (expected: Negative) to verify that DistilBERT predictions align with real user expectations.
- Route planning validation entering a destination beyond the vehicle's current battery range and confirming that the system correctly inserts charging stops at appropriate waypoints along the route.
- Usability validation confirming that new users can complete core tasks (register, find a station, plan a route, submit a review) without external guidance, based on the intuitive screen design.

2.8 Usability Testing

Usability testing evaluates how easily and efficiently users can interact with the Smart EV Charging and Navigation Management System. The objective is to ensure that all three user types EV Owner, Station Owner, and Administrator can navigate the application and complete their tasks without confusion or errors.

The following usability aspects were evaluated:

- Navigation clarity all major screens are reachable within 2-3 taps from the home screen. Role-based dashboards are clearly separated so users immediately know where they are.
- Map interaction the nearby stations map with GPS detection, filter panel (charger type, max price, search bar), and station detail bottom sheet were tested for ease of use on mobile screen sizes.
- Route planning flow the sequence of selecting start location, entering destination, choosing vehicle profile, and setting battery percentage was evaluated for logical progression and minimal user input.
- Review submission the star rating + text input + DistilBERT sentiment feedback flow was tested to ensure users understand the process and receive clear confirmation of successful submission.

- Admin panel usability the admin's ability to view, filter, verify, activate/deactivate, and delete user and station records was tested for clarity and speed.
- Usability testing was carried out by team members acting as representative users for each role. Observations were documented and minor UI adjustments were made based on feedback. No external users were available for formal usability sessions; this is identified as a future improvement.

2.9 Module / Unit Testing

Unit testing verifies that each individual function, service, and screen in the system works correctly in isolation. For the Smart EV Charging and Navigation Management System, the following units were tested independently:

- Authentication functions register(), login(), logout(), and sendPasswordResetEmail() in auth_store.dart were tested with valid and invalid inputs to confirm correct Firebase responses and error handling.
- Route optimization functions the A* shortest path, A* fastest path, and Yen's K-Shortest Paths (k=4) were tested with sample graph nodes to verify correct path selection and scoring.
- EV energy calculation consumptionRate, energyKwh, remainingRangeKm, arrivalBattery%, and stopsNeeded calculations were verified against manual calculations for known vehicle profiles and distances.
- Sentiment API the FastAPI /analyze_review endpoint was tested with sample review texts to confirm Positive, Neutral, and Negative classification with confidence scores.
- Firestore CRUD operations station_store.dart, review_store.dart, vehicle_store.dart, and audit_store.dart functions were tested for correct read, write, update, and delete behaviour.
- Station filters the type filter (AC/DC), price filter (max price slider), and name/area search were unit tested in ev_owner_nearby_stations_page.dart to confirm correct station list results.

2.10 Integration Testing

Integration testing verifies that the individual modules of the system communicate and exchange data correctly when combined. The following integration points were tested:

- Flutter: Firebase user registration data flows from the auth form through auth_store.dart to Firebase Authentication and Firestore user collection without data loss or format errors.
- Flutter: FastAPI ML Backend when a review is submitted, the review text is correctly sent as a POST request to the /analyze_review endpoint, and the returned sentiment label and confidence score are correctly stored in Firestore and displayed on the admin review panel.
- Flutter: OSRM API the route planner page correctly calls DirectionsService.fetchAlternatives(), receives polyline data, converts it into a graph, and passes it to RouteOptimizationService for A* and Yen's processing.

- Vehicle profile: Route planning the selected vehicle's batteryKwh and rangeKm are correctly consumed by the route evaluation function to calculate energy usage, arrival battery percentage, and required charging stops.
- Station data: Cost comparison station price data from Firestore is correctly used in the cost comparison page to calculate and rank estimated charging costs.
- Admin audit log: Station verification when admin approves or rejects a station, the audit log correctly records the action with timestamp, admin ID, and station ID.

2.11 System Testing

System testing validates the complete Smart EV Charging and Navigation Management System as a whole including the Flutter mobile app, Firebase backend, OSRM routing API, and FastAPI ML service to confirm that all components work together and all functional requirements are satisfied end-to-end.

The following end-to-end scenarios were included in system testing:

- Full EV Owner flow: Register → Login → Add vehicle profile → Search nearby stations → Filter by AC charger → View station details and amenities → Plan route with battery level → View Efficient / Fastest / Shortest routes → Submit review → Logout.
- Route planning with charging stop: Set destination beyond current battery range → Confirm system inserts nearest verified open station as a waypoint → Verify FR-17 warning appears when no station is in range.
- Station Owner flow: Register as station owner → Submit new station (pending admin approval) → Edit station details → Update status to Closed → View session history and usage report.
- Admin flow: Login as admin → Approve pending station → View flagged review from ML model → Remove inappropriate review → Deactivate a user account → View audit log entry.
- ML sentiment flow: Submit a clearly positive review → Confirm DistilBERT returns Positive label → Submit an abusive review → Confirm it is flagged and appears in admin review panel.

2.12 Acceptance Testing

Acceptance testing confirms that the Smart EV Charging and Navigation Management System satisfies the business goals and user needs defined at the start of the project. It is the final level of testing before submission and focuses on real-world usability and requirement coverage.

Acceptance criteria for this project:

- All 51 functional requirements (FR-1 to FR-52, excluding FR-32 which was descope) are implemented and demonstrate correct behaviour during demo.
- The three user roles EV Owner, Station Owner, and Admin can each complete their core workflows without errors or system crashes.
- The DistilBERT sentiment analysis model correctly classifies EV charging-related reviews during live demonstration with test inputs.

- Route planning with OSRM integration returns meaningful Efficient, Fastest, and Shortest routes and correctly places charging stops when battery range is insufficient.
- The admin portal correctly displays ML-flagged reviews and allows removal with audit logging.
- The application runs correctly on an Android device/emulator with the FastAPI ML server running locally on port 8000 and the Flutter app connecting via 10.0.2.2:8000.

Acceptance testing was carried out by the project team simulating supervisor review and panel presentation scenarios. All high-priority acceptance criteria were met. FR-52 (filter by amenities) is marked as In Progress and documented in the RTM.

2.13 Manual Testing

All testing for the Smart EV Charging and Navigation Management System was conducted manually. No automated testing framework (such as Flutter test, Mockito, or CI/CD pipeline scripts) was used in this version of the project. Manual testing was selected because it is appropriate for the scale of a student capstone project and allows testers to observe the complete user interface behaviour during execution.

The manual testing approach followed these steps:

- Each test case was prepared in advance with a defined test ID, title, module name, test data, preconditions, step-by-step actions, and expected result – as shown in Section 7.14.
- Testers followed the step-by-step actions listed in each test case and observed the system response on an Android emulator (Android Studio) and a physical Android device.
- Actual results were recorded and compared with expected results. Status was marked as Pass or Fail.
- Failed test cases were reported to the responsible developer, fixed, and retested in the next cycle.
- The FastAPI ML backend was started manually on a local machine using the command: `uvicorn main:app --reload --host 0.0.0.0 --port 8000`, and the Flutter app connected to it via the Android emulator bridge address 10.0.2.2:8000.

A total of 10 module-level test cases and 40+ detailed sub-test cases were designed and executed manually covering all functional modules. Results are recorded in Section 7.14.

2.14 Test Cases

Write the test cases of your project as per module wise.

Table 1: Test Case-1 (Module 01)

Test Id:	TC-01	Test Case Designed by:	Zoya Khalid
Test Case Title:	User authentication flow works correctly	Test Case Executed by:	Sarwat Naveed

Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026
Test Data:	Email: test@gmail.com, Password: test@1234	Priority:	High
Precondition:		User has app access and is not logged in.	
Steps /Action		System Response	
1. Open EV Charging Application 2. System displays “Welcome Screen” 3. Click on “Register” option 4. Enter valid email and password 5. Click on “Register” button 6. Logout from the application 7. Click on “Login” option 8. Enter registered email and password 9. Click on “Login” button		1. Application launches successfully. 2. Welcome Screen is displayed. 3. Registration page opens. 4. System validates the entered email and password. 5. User account is created successfully. 6. User is logged out successfully. 7. Login page opens. 8. System validates the registered credentials. 9. User is authenticated successfully and redirected to the dashboard.	
Expected Result:		User account should be created successfully, login should work with valid credentials, and only authorized users should be allowed to access the system.	
Actual Result:		User account was created successfully. The registered user logged in successfully with valid credentials and gained access to the system	
Status:		Pass	

Table 2: TC001-M1-A

Test Id:	TC-001-M1-A	Test Case Designed by:	Zoya Khalid
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Test Case Title:	User registration with valid email and password	Test Case Executed by:	07-07-2026
Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026
Test Data:	Email: newuser@gmail.com, Password: Strong@123	Priority:	High
Precondition:		User is not already registered in the system and has access to the application.	
Steps /Action		System Response	
1. Open EV Charging Application 2. System displays “Welcome Screen” 3. Click on “Register” option 4. System displays registration form 5. Enter valid email and password 6. Click on “Register” button		1. Application launches successfully. 2. Welcome Screen is displayed. 3. Registration page opens. 4. Registration form is displayed. 5. System validates the entered email and password. 6. User account is created successfully, credentials are stored securely.	
Expected Result:		The system should successfully create a new user account and store user credentials securely.	
Actual Result:		The user account was created successfully with valid credentials, and the registration process completed without errors.	
Status:		Pass	

Table 3: TC001-M1-B

Test Id:	TC-001-M1-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	User registration with invalid email format	Test Case Executed by:	Sarwat Naveed
Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026

Test Data:	Email: usergmail.com, Password: Strong@123	Priority:	High
Precondition:		User is on registration screen	
Steps /Action		System Response	
1. Open EV Charging Application 2. Click on “Register” 3. Enter invalid email format 4. Enter valid password 5. Click on “Register”		1. Application launches successfully. 2. Registration page is displayed. 3. System detects the invalid email format. 4. Password is accepted as it meets the required criteria. 5. Registration request is rejected and an "Invalid email format" validation message is displayed.	
Expected Result:		System should reject invalid email and display appropriate error message.	
Actual Result:		The system rejected the invalid email address and displayed an appropriate validation message. No user account was created.	
Status:		Pass	

Table 4: TC002-M1-A

Test Id:	TC-002-M1-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Password recovery using registered email	Test Case Executed by:	Sarwat Naveed
Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026
Test Data:	Email: usergmail.com, Password: Strong@123	Priority:	High
Precondition:		User is registered.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Click on “Login” 3. Click on “Forgot Password” 4. Enter registered email 5. Click submit		1. Application launches successfully. 2. Login page is displayed. 3. Forgot Password page opens successfully. 4. System validates the registered email address.	

	5. Password reset request is processed successfully and a password reset link/instructions are sent to the registered email.
Expected Result:	Password recovery process should start successfully.
Actual Result:	The password recovery process was initiated successfully, and the reset instructions were sent to the registered email address.
Status:	Pass

Table 5: TC002-M1-B

Test Id:	TC-002-M1-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	Password recovery with unregistered email	Test Case Executed by:	Sarwat Naveed
Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026
Test Data:	Email: usergmail.com, Password: Strong@123	Priority:	High
Precondition:	User is registered.		
Steps /Action	System Response		
1. Open forgot password 2. Enter unregistered email 3. Click submit	1. Forgot Password page opens successfully. 2. System checks the entered email against the registered user database. 3. System displays an "Email not found" error message and does not initiate the password recovery process.		
Expected Result:	System should display "Email not found" error.		
Actual Result:	The system displayed an "Email not found" message, and no password reset request was generated.		
Status:	Pass		

Table 6: TC003-M1-A

Test Id:	TC-003-M1-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Secure login with valid credentials	Test Case Executed by:	Sarwat Naveed
Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026
Test Data:	Email: usergmail.com, Password: Strong@123	Priority:	High
Precondition:		User is registered.	
Steps /Action		System Response	
1. Open application 2. Click login 3. Enter valid credentials 4. Click login		1. Application launches successfully. 2. Login page is displayed. 3. System validates the entered email and password. 4. User is authenticated successfully and redirected to the dashboard.	
Expected Result:		User should be logged in securely.	
Actual Result:		The user logged in successfully using valid credentials and gained secure access to the application.	
Status:		Pass	

Table 7: TC003-M1-B

Test Id:	TC-003-M1-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	Login attempt with invalid credentials	Test Case Executed by:	Sarwat Naveed
Module Name:	User Registration & Authentication	Test Case Execution Date:	07-07-2026
Test Data:	Email: usergmail.com, Password: Strong@123	Priority:	High
Precondition:		User is registered.	

Steps /Action	System Response
1. Enter wrong credentials 2. Click login	1. System validates the entered credentials. 2. Authentication fails, and an "Invalid email or password" error message is displayed.
Expected Result:	System should deny access.
Actual Result:	The system denied access for invalid credentials and displayed the appropriate error message.
Status:	Pass

Table 8: Test Case-2 (Module 02)

Test Id:	TC-02	Test Case Designed by:	Zoya Khalid
Test Case Title:	Charging station locator works correctly	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026
Test Data:	GPS enabled, current location available, charger type: Fast Charger, price range: PKR 80–100	Priority:	High
Precondition:		User is logged in, internet is available, and GPS/location service is enabled.	
Steps /Action		System Response	

1. Open EV Charging Application 2. Login to the application 3. Open “Charging Station Locator” 4. Allow GPS/location access 5. View charging stations on map 6. Select a charging station 7. Apply filters for location, charger compatibility, or price	1. EV Charging Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator screen opens. 4. System requests and receives GPS/location permission. 5. Nearby charging stations are displayed on the map. 6. Selected charging station details are loaded successfully. 7. System filters stations according to location, charger compatibility, and price.
Expected Result:	The system should display nearby charging stations using GPS and map integration, show complete station details, and allow filtering by location, charger compatibility, and price.
Actual Result:	Nearby charging stations were displayed correctly with complete station details, and filters worked as expected.
Status:	Pass

Table 9: TC002-M2-A

Test Id:	TC001-M2-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Display nearby charging stations with GPS enabled	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026
Test Data:	GPS Location Enabled	Priority:	High
Precondition:		User is logged in and GPS/location service is enabled.	
Steps /Action		System Response	

1. Open EV Charging Application 2. Login to the application 3. Open “Charging Station Locator” 4. Allow GPS/location access 5. Click on “Find Charging Stations”	1. Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator opens. 4. GPS permission is granted successfully. 5. System retrieves the user's current location and displays nearby charging stations on the map.
Expected Result:	Nearby charging stations should be displayed on the map based on the user’s current location.
Actual Result:	Nearby charging stations were displayed accurately based on the user's current GPS location.
Status:	Pass

Table 10: TC002-M2-B

Test Id:	TC001-M2-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	Access charging station locator with GPS disabled	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026
Test Data:	GPS Disabled	Priority:	High
Precondition:	User is logged in but GPS/location service is disabled.		
Steps /Action	System Response		
1. Open EV Charging Application 2. Login to the application 3. Open “Charging Station Locator” 4. Disable GPS/location service 5. Click on “Find Charging Stations”	1. Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator opens. 4. System detects that GPS is disabled. 5. System prompts the user to enable location services before continuing.		
Expected Result:	System should prompt the user to enable location services before searching nearby charging stations.		
Actual Result:	The application displayed a message requesting the user to enable GPS before searching for charging stations.		
Status:	Pass		

Table 11: TC002-M2-A

Test Id:	TC002-M2-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	View complete charging station details	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026

Test Data:	GPS enabled, approved station available on map	Priority:	High
Precondition:		User is logged in and charging stations are available on the map.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Login to the system 3. Open “Charging Station Locator” 4. Select a charging station from the map		1. Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator opens. 4. System loads the selected charging station details, including station name, charger type, charging cost, availability, and location.	
Expected Result:		System should display complete station details including station name, charger type, cost per unit, status, location, and available information.	
Actual Result:		Complete charging station information was displayed correctly without missing required details.	
Status:		Pass	

Table 12: TC002-M2-B

Test Id:	TC002-M2-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	View station details when data is incomplete	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026
Test Data:	Station record with missing optional details	Priority:	High
Precondition:		User is logged in and at least one station has incomplete optional information.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Login to the application 3. Open “Charging Station Locator” 4. Select the station with incomplete data		1. Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator opens. 4. System displays available station information and handles missing optional data without affecting application performance	
Expected Result:		System should handle missing data gracefully and display available station information without crashing.	
Actual Result:		Available station details were displayed successfully, and the application remained stable despite incomplete data.	
Status:		Pass	

Table 13: TC003-M2-A

Test Id:	TC-003-M2-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Filter stations based on compatibility and price	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026
Test Data:	GPS Enabled	Priority:	High
Precondition:		User is logged in.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Login to the application 3. Open “Charging Station Locator” 4. Apply filter (e.g., Fast charger / Low price)		1. Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator opens. 4. System applies the selected compatibility and price filters and refreshes the station list.	
Expected Result:		Only stations matching filter criteria should be displayed.	
Actual Result:		Only charging stations matching the selected filter criteria were displayed.	
Status:		Pass	

Table 14: TC003-M2-B

Test Id:	TC-003-M2-B	Test Case Designed by:	Zoya Khalid
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Test Case Title:	Apply invalid filter combination	Test Case Executed by:	Sarwat Naveed
Module Name:	Charging Station Locator	Test Case Execution Date:	07-07-2026
Test Data:	GPS Enabled	Priority:	High
Precondition:		User is logged in.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Login to the application 3. Open “Charging Station Locator” 4. Apply incompatible filters		1. Application launches successfully. 2. User logs in successfully. 3. Charging Station Locator opens. 4. System processes the selected filter combination and finds no matching charging stations	
Expected Result:		System should display no results message.	
Actual Result:		The application displayed a "No charging stations found" message without any errors.	
Status:		Pass	

Table 15: Test Case-3 (Module 03)

Test Id:	TC-03	Test Case Designed by:	Zoya Khalid
Test Case Title:	Route planning works correctly	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Battery Percentage: 35%, Selected Vehicle: MG ZS EV, Battery Capacity: 50 kWh, Range: 320 km	Priority:	High
Precondition:		User is logged in and a vehicle is already selected from Vehicle Management module.	
Steps /Action		System Response	

1. Open EV Charging application 2. Login to the application 3. Open “Route Planning” screen 4. Manually enter current battery percentage 5. Select saved vehicle details 6. Start route planning 7. View nearest charging stations 8. Test condition where no station is available in range	1. EV Charging Application launches successfully. 2. User logs into the application successfully. 3. Route Planning screen is displayed. 4. System accepts the entered battery percentage. 5. Selected vehicle details are loaded successfully. 6. Route planning process starts. 7. System calculates the route and displays the nearest charging stations within range. 8. If no charging station is available, the system displays an alert message.
Expected Result:	The system should calculate remaining travel distance using battery percentage and selected vehicle details, suggest nearest charging stations within range, and display an alert if no station is available.
Actual Result:	The route was calculated successfully based on the battery percentage and selected vehicle. Nearby charging stations were displayed correctly, and an alert appeared when no station was available within range.
Status:	Pass

Table 16: TC001-M3-A

Test Id:	TC-001-M1-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Enter valid battery percentage	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Battery Percentage: 35%, Vehicle selected	Priority:	High
Precondition:		User is logged in and a vehicle is already selected from Vehicle Management module.	
Steps /Action		System Response	

1. Open EV Charging application 2. Login to the application 3. Open “Route Planning” screen 4. Enter battery percentage (e.g., 35%) 5. Submit	1. Application launches successfully. 2. User logs in successfully. 3. Route Planning screen opens. 4. System validates and accepts the entered battery percentage. 5. Battery value is saved successfully for route calculation.
Expected Result:	Battery input should be accepted.
Actual Result:	The valid battery percentage was accepted successfully and used for route planning.
Status:	Pass

Table 17: TC001-M3-B

Test Id:	TC-001-M3-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	Enter invalid battery percentage	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Battery Percentage: 150%, Vehicle selected	Priority:	High
Precondition:		User is logged in and a vehicle is already selected from Vehicle Management module.	
Steps /Action		System Response	
1. Open EV Charging application 2. Login to the application 3. Open “Route Planning” screen 4. Enter battery value (e.g., 150%) 5. Submit		1. Application launches successfully. 2. User logs in successfully. 3. Route Planning screen opens. 4. System detects that the battery percentage is outside the valid range (0–100%). 5. System displays a validation error and rejects the entered value.	
Expected Result:		System should reject invalid battery value.	
Actual Result:		The invalid battery percentage was rejected, and an appropriate validation message was displayed.	
Status:		Pass	

Table 18: TC002-M3-A

Test Id:	TC-002-M3-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Vehicle-based calculation	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle selected	Priority:	High
Precondition:		User is logged in to the system.	
Steps /Action		System Response	
1. Open EV Charging application 2. Login to the application 3. Open “Route Planning” screen 4. Select saved vehicle 5. Enter destination 6. Click plan route		1. Application launches successfully. 2. User logs in successfully. 3. Route Planning screen opens. 4. Selected vehicle information is loaded successfully. 5. Destination is accepted successfully. 6. System calculates the route and estimated travel distance using vehicle specifications.	
Expected Result:		Distance should be calculated correctly.	
Actual Result:		The route distance and battery consumption were calculated accurately based on the selected vehicle.	
Status:		Pass	

Table 19:TC002-M3-B

Test Id:	TC-002-M3-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	Attempt route planning without selecting a vehicle	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle not selected, destination entered	Priority:	High
Precondition:		User is logged in to the system but no vehicle profile is selected.	
Steps /Action		System Response	

1. Open EV Charging Application 2. Login to the application 3. Open “Route Planning” screen 4. Do not select a vehicle 5. Enter destination 6. Click “Plan Route”	1. Application launches successfully. 2. User logs in successfully. 3. Route Planning screen opens. 4. System detects that no vehicle has been selected. 5. Destination is accepted temporarily. 6. System displays an error asking the user to select a vehicle before planning the route.
Expected Result:	System should ask the user to select a saved vehicle before calculating route distance and charging stops.
Actual Result:	The application prevented route planning until a vehicle was selected and displayed the appropriate error message.
Status:	Pass

Table 20: TC003-M3-A

Test Id:	TC-003-M3-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Suggest stations	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle selected	Priority:	High
Precondition:		User is logged in to the system.	
Steps /Action		System Response	
1. Open EV Charging application 2. Login to the application 3. Open “Route Planning” screen 4. Enter low battery 5. Plan route		1. Application launches successfully. 2. User logs in successfully. 3. Route Planning screen opens. 4. System detects the low battery percentage. 5. System calculates the route and recommends nearby charging stations within the available battery range.	
Expected Result:		Stations within range should be suggested.	
Actual Result:		Nearby charging stations within the vehicle's remaining battery range were displayed successfully.	
Status:		Pass	

Table 21: TC003-M3-B

Test Id:	TC-003-M3-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	No stations available	Test Case Executed by:	Sarwat Naveed
Module Name:	Route Planning	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle selected	Priority:	High
Precondition:		User is logged in to the system.	
Steps /Action		System Response	
1. Open EV Charging application 2. Login to the application 3. Open “Route Planning” screen 4. Enter destination 5. Plan route		1. Application launches successfully. 2. User logs in successfully. 3. Route Planning screen opens. 4. Destination is accepted successfully. 5. System searches for charging stations but finds none within the reachable range and displays an alert.	
Expected Result:		System should alert user that no station found in given range.	
Actual Result:		The application displayed a "No charging station available within range" message correctly.	
Status:		Pass	

Table 22:Test Case-4 (Module 04)

Test Id:	TC-04	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Vehicle Management module works correctly	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle Name: MG ZS EV, Model: 2023, Battery: 50 kWh, Range: 320 km	Priority:	High

Precondition:	User is logged in to the EV Owner account.
Steps /Action	System Response
1. Open EV Charging Application 2. Login as EV Owner 3. Open “Vehicle Management” 4. Add, view, update, and delete vehicle profile 5. Select a vehicle for route planning	1. Application launches successfully. 2. User logs in as an EV Owner successfully. 3. Vehicle Management module opens successfully. 4. System performs add, view, update, and delete operations on vehicle profiles successfully. 5. Selected vehicle is linked successfully with the Route Planning module.
Expected Result:	System should allow the user to add, view, update, delete, and select vehicle profiles correctly.
Actual Result:	All vehicle management operations completed successfully, and the selected vehicle was available for route planning.
Status:	Pass

Table 23: TC001-M4-A

Test Id:	TC-001-M4-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Add Vehicle Profile with Valid Data	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle Name: Tesla, Model: Model S, Battery: 75 kWh, Range: 400 km	Priority:	High
Precondition:	User is logged into the system Vehicle Management module is accessible.		
Steps / Action		System Response	
1. The user logs into the system with valid credentials.		1. User is authenticated successfully. 2. Vehicle Management module opens. 3. Add Vehicle form is displayed.	

2. The user navigates to the Vehicle Management module. 3. The user clicks on the "Add Vehicle" button. 4. The user enters valid vehicle details (Name: Tesla, Model: Model S, Battery: 75, Range: 400). 5. The user clicks on the "Save" button.	4. System validates the entered vehicle information. 5. Vehicle profile is saved successfully and displayed in the vehicle list.
Expected Result:	Vehicle profile (Tesla Model S) is saved successfully and displayed in the vehicle list.
Actual Result:	The vehicle profile was created successfully and appeared in the vehicle list.
Status:	Pass

Table 24:TC001-M4-B

Test Id:	TC-001-M4-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Add Vehicle Profile with Missing Required Fields	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle Name: (empty), Battery: (empty)	Priority:	High
Precondition:	User is logged into the system. Vehicle Management module is accessible.		
Steps / Action		System Response	
1. The user navigates to the Vehicle Management module. 2. The user clicks on the "Add Vehicle" button. 3. The user leaves the Vehicle Name and Battery Capacity fields empty. 4. The user clicks on the "Save" button.		1. Vehicle Management module opens. 2. Add Vehicle form is displayed. 3. System detects missing required fields. 4. System displays validation messages and prevents saving the record.	

Expected Result:	System displays a validation error for missing required fields and does not save the vehicle profile.
Actual Result:	The application displayed validation errors and did not save the incomplete vehicle profile.
Status:	Pass

Table 25:TC002-M4-A

Test Id:	TC-002-M4-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	View All Saved Vehicle Profiles	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	User account with 2 saved vehicles: Tesla Model S, Nissan Leaf	Priority:	Medium
Precondition:	User is logged into the system. At least one vehicle profile has been added (FR-18 satisfied).		
Steps / Action		System Response	
The user logs into the system with valid credentials. The user navigates to the Vehicle Management module. The user clicks on the "My Vehicles" section. The user views the list of all saved vehicle profiles. The user selects a vehicle to view its details.		1. User is authenticated successfully. 2. Vehicle Management module opens. 3. System displays the "My Vehicles" section. 4. All saved vehicle profiles are loaded successfully. 5. Complete details of the selected vehicle are displayed.	
Expected Result:	All saved vehicle profiles are displayed. Correct vehicle details (name, model, battery, range) are shown for the selected vehicle.		
Actual Result:	All saved vehicle profiles and their details were displayed correctly.		
Status:	Pass		

Table 26:TC002-M4-B

Test Id:	TC002-M4-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	View Vehicle Profiles When No Vehicle Exists	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	New user account with no saved vehicles	Priority:	Medium
Precondition:	1. User is logged into the system. 2. No vehicle profiles have been added to the account.		
Steps / Action		System Response	
1. The user logs into the system as a new user. 2. The user navigates to the Vehicle Management module. 3. The user opens the "My Vehicles" section. 4. The user attempts to view the vehicle list.		1. User logs in successfully. 2. Vehicle Management module opens. 3. "My Vehicles" section is displayed. 4. System checks for saved vehicles and displays a "No vehicles found" message.	
Expected Result:	System displays a 'No vehicles found' message when the vehicle list is empty.		
Actual Result:	The application displayed an appropriate message indicating that no vehicles were available.		

Status:	Pass
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Table 27:TC003-M4-A

Test Id:	TC003-M4-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Edit Vehicle Profile with Valid Data	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Existing: Tesla Model S, Updated Model: Model X, Updated Battery: 100 kWh	Priority:	Medium
Precondition:	User is logged into the system. At least one vehicle profile exists in the account		
Steps / Action		System Response	
1. The user navigates to the Vehicle Management module. 2. The user selects the vehicle profile to edit. 3. The user clicks on the "Edit" button. 4. The user updates the vehicle model to "Model X" and battery capacity to 100. 5. The user clicks on the "Save" button.		1. Vehicle Management module opens. 2. Selected vehicle details are loaded. 3. Edit form opens successfully. 4. Updated vehicle information is validated successfully. 5. Vehicle profile is updated and saved successfully.	
Expected Result:	Vehicle profile is updated successfully with Model X and 100 kWh battery capacity.		
Actual Result:	The vehicle profile was updated successfully with the new information.		
Status:	Pass		

Table 28:TC003-M4-B

Test Id:	TC003-M4-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Edit Vehicle Profile with Invalid Battery Value	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle Name: Tesla, Battery: -50	Priority:	Medium
Precondition:	User is logged into the system. Vehicle profile exists in the account. Edit form is open.		
Steps / Action		System Response	
1. The user navigates to the Vehicle Management module. 2. The user selects a vehicle and clicks "Edit". 3. The user enters an invalid negative value in the Battery field (-50). 4. The user clicks on the "Save" button.		1. Vehicle Management module opens. 2. Edit form opens successfully. 3. System detects an invalid battery value. 4. System displays a validation error and rejects the update.	
Expected Result:	System displays a validation error and rejects the negative battery value of -50.		
Actual Result:	The invalid battery value was rejected, and the vehicle profile remained unchanged.		
Status:	Pass		

Table 29:TC004-M4-A

Test Id:	TC004-M4-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Delete Vehicle Profile with Confirmation	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026

Test Data:	Existing Vehicle: Tesla Model S (Vehicle ID: V001)	Priority:	Medium
Precondition:	User is logged into the system. Vehicle profile exists in the account		
Steps / Action		System Response	
1. The user navigates to the Vehicle Management module. 2. The user selects the vehicle profile to delete. 3. The user clicks on the "Delete" button. 4. The system shows a confirmation dialog. 5. The user clicks "Confirm" to proceed with deletion.		1. Vehicle Management module opens. 2. Selected vehicle is identified successfully. 3. Delete request is initiated. 4. Confirmation dialog is displayed. 5. After confirmation, the vehicle profile is deleted successfully.	
Expected Result:	Vehicle profile is deleted successfully and removed from the vehicle management list.		
Actual Result:	The selected vehicle profile was deleted successfully and removed from the list.		
Status:	Pass		

Table 30:TC004-M4-B

Test Id:	TC004-M4-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Cancel Vehicle Deletion at Confirmation	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Existing Vehicle: Nissan Leaf (Vehicle ID: V002)	Priority:	Medium

Precondition:	User is logged into the system. Vehicle profile exists. Delete confirmation dialog is displayed.	
Steps / Action		System Response
1. The user navigates to the Vehicle Management module. 2. The user selects a vehicle profile. 3. The user clicks on the "Delete" button. 4. The system displays a confirmation dialog. 5. The user clicks "Cancel" to abort the deletion.		1. Vehicle Management module opens. 2. Selected vehicle is identified successfully. 3. Delete request is initiated. 4. Confirmation dialog is displayed. 5. System cancels the deletion request and keeps the vehicle profile unchanged.
Expected Result:	Vehicle deletion is cancelled and the profile remains intact in the system.	
Actual Result:	Vehicle deletion was cancelled successfully, and the vehicle profile remained available.	
Status:	Pass	

Table 31:TC005-M4-A

Test Id:	TC005-M4-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Select Vehicle for Route Planning Successfully	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	Vehicles: Tesla Model S, Nissan Leaf; Route: Lahore to Islamabad	Priority:	High
Precondition:	User is logged into the system. Multiple vehicle profiles exist. Route Planning module is accessible.		
Steps / Action		System Response	

1. The user navigates to the Route Planning module. 2. The user clicks on the "Select Vehicle" option. 3. The system displays the list of saved vehicle profiles. 4. The user selects "Tesla Model S" from the list. 5. The user confirms the vehicle selection.	1. Route Planning module opens. 2. Select Vehicle option is displayed. 3. System loads all saved vehicle profiles. 4. Selected vehicle is highlighted successfully. 5. Vehicle selection is confirmed and linked with the route planning session.
Expected Result:	Tesla Model S is selected successfully and linked to the route planning session.
Actual Result:	The selected vehicle was linked successfully with the route planning process.
Status:	Pass

Table 32:TC005-M4-B

Test Id:	TC005-M4-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Attempt Route Planning Without Selecting a Vehicle	Test Case Executed by:	Waleed
Module Name:	Vehicle Management	Test Case Execution Date:	07-07-2026
Test Data:	No vehicle selected; Route: Lahore to Islamabad	Priority:	High
Precondition:	User is logged into the system. Route Planning module is accessible. No vehicle is selected.		
Steps / Action		System Response	
1. The user navigates to the Route Planning module. 2. The user does not select any vehicle profile.		1. Route Planning module opens. 2. No vehicle is selected. 3. User attempts to start route planning.	

3. The user directly attempts to start route planning without selecting a vehicle. 4. The user clicks the "Plan Route" button.	4. System displays an error requesting the user to select a vehicle before planning the route.
Expected Result:	System displays an error message prompting the user to select a vehicle before planning a route.
Actual Result:	The application prevented route planning and displayed the required validation message.
Status:	Pass

Table 33: Test Case-5(Module 05)

Test Id:	TC-05	Test Case Designed by:	Zoya Khalid
Test Case Title:	Cost comparison and travel expense estimation work correctly	Test Case Executed by:	Waleed
Module Name:	Cost Comparison and Estimation	Test Case Execution Date:	07-07-2026
Test Data:	Station A: PKR 85/kWh, Station B: PKR 95/kWh	Priority:	High
Precondition:		User is logged in and station cost data is available.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Login to the application 3. Open station details or route result 4. Select multiple charging stations 5. Compare charging rates		1. EV Charging Application launches successfully. 2. User logs into the application successfully. 3. System opens the Station Details/Route Result screen. 4. System loads the selected charging stations for comparison. 5. System compares the charging rates and displays the estimated charging cost and travel expense for each station.	
Expected Result:		System should display charging rates from different stations and help the user compare the estimated charging cost.	
Actual Result:		The application successfully compared the charging rates of different stations and displayed the estimated charging cost and travel expense accurately.	
Status:		Pass	

Table 34: TC001-M5-A

Test Id:	TC001-M5-A	Test Case Designed by:	Zoya Khalid
Test Case Title:	Compare charging rates between stations	Test Case Executed by:	Waleed
Module Name:	Cost Comparison and Estimation	Test Case Execution Date:	07-07-2026
Test Data:	Select any option from given stations	Priority:	High
Precondition:		User is logged in and charging rates from different stations are available.	
Steps /Action		System Response	
1. Open application 2. Login to the application 3. Open “Cost Comparison and Estimation” 4. View charging rates from different stations 5. Compare station prices and distance		1. Application launches successfully. 2. User logs into the application successfully. 3. Cost Comparison and Estimation module opens successfully. 4. System retrieves and displays charging rates, distance, and station details. 5. System compares the selected stations and calculates the estimated charging cost and travel expense.	
Expected Result:		The system should display charging rates from different stations and help the user estimate travel expenses before starting the journey.	
Actual Result:		Charging rates, station distance, and estimated travel expenses were displayed correctly, allowing the user to compare different charging stations.	
Status:		Pass	

Table 35: TC001-M5-B

Test Id:	TC001-M5-B	Test Case Designed by:	Zoya Khalid
Test Case Title:	Compare stations with missing cost data	Test Case Executed by:	Waleed

Module Name:	Cost Comparison and Estimation	Test Case Execution Date:	07-07-2026
Test Data:	Select any option from given stations	Priority:	High
Precondition:		User is logged in.	
Steps /Action		System Response	
1. Open application 2. Login to the application 3. Open “Cost Comparison and Estimation” 4. Select station with missing cost 5. Compare station prices and distance		1. Application launches successfully. 2. User logs into the application successfully. 3. Cost Comparison and Estimation module opens successfully. 4. System detects that the selected station has missing cost information. 5. System displays the available station information and shows an appropriate message such as "Cost information unavailable" without affecting the comparison process.	
Expected Result:		System should handle missing data properly.	
Actual Result:		The application handled the missing cost data correctly, displayed the available information, and continued functioning without errors.	
Status:		Pass	

Table 36:Test Case-6(Module 06)

Test Id:	TC-06	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Charging Station Owner Portal works correctly	Test Case Executed by:	Zoya Khalid
Module Name:	Charging Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Owner account, station details, charger type, price, availability, operating hours, documents	Priority:	High
Precondition:		Station owner has access to the application and internet is available.	
Steps /Action		System Response	
1. Register or login as station owner 2. Register a charging station		1. System verifies credentials and loads station owner dashboard.	

3. Update station details 4. Upload verification documents 5. Save changes	2. Station registration form opens with all required fields. 3. System validates inputs and saves station details. 4. Availability status updates in real time via Firebase. 5. System confirms all changes saved successfully.
Expected Result:	System should allow station owner registration, station registration, station detail updates, availability updates, operating hours updates, and verification document upload.
Actual Result:	System allowed station owner login, station registration, detail updates, and availability status change successfully.
Status:	Pass

Table 37:TC001-M6-A

Test Id:	TC001-M6-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Register New Charging Station with Valid Data	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Owner: Ali Khan, Station Name: GreenCharge Station, Charger Type: Fast Charger	Priority:	High
Precondition:	Station owner is logged in and has access to the registration portal.		
Steps / Action		System Response	
1. The user logs into the EV Owner Portal using valid credentials. 2. The user navigates to the profile section. 3. The user opens the profile edit screen. 4. The system enables editable fields for name and email.		1. Login screen authenticates credentials and loads station owner dashboard. 2. Add New Station form opens with all required input fields. 3. System accepts all entered values: name, charger type, price, and map pin location. 4. Amenities checkboxes are selectable. Wi-Fi and Parking marked successfully.	

5. The user updates the name and email with valid values.	5. System submits station and displays confirmation: "Station submitted for admin approval."
6. The user clicks on the "Save" button	
Expected Result:	Charging station is registered successfully and submitted for admin approval.
Actual Result:	Charging station GreenCharge registered successfully and submitted for admin approval. Confirmation message displayed.
Status:	Pass

Table 38:TC001-M6-B

Test Id:	TC001-M6-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Register Charging Station with Missing Details	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Missing Station Name, Charger Type: Fast Charger	Priority:	High
Precondition:	Owner logged into portal registration form accessible.		
Steps / Action		System Response	
1. The station owner navigates to the “Owner Portal”. 2. The station owner opens the registration form. 3. The station owner leaves required fields empty. 4. The station owner enters incomplete data. 5. The station owner clicks on “Submit”.		1. Station owner dashboard loads. 2. Registration form opens. 3. Required fields (Station Name) left empty system does not auto-fill. 4. Partial data entered in other fields. 5. System detects empty required field on submit. Validation error displayed: "Station name is required."	
Expected Result:	System displays a validation error and does not allow registration with missing required fields.		

Actual Result:	System displayed validation error for missing station name and did not allow registration to proceed.
Status:	Pass

Table 39:TC002-M6-A

Test Id:	TC002-M6-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Station Details with Valid Data	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station Name: Green Charge, Charger Type: Fast, Ultra-Fast Charger	Priority:	High
Precondition:	1. Station should be existing in the system 2. user should be authorized owner.		
Steps / Action		System Response	
1. The station owner navigates to the “Station Details” page. 2. The station owner selects an existing charging station. 3. The station owner updates the charger type information. 4. The station owner modifies the station details with valid data. 5. The station owner clicks on the “Save Changes” button.		1. Station owner dashboard loads after login. 2. My Stations list display all registered stations. 3. Station detail page opens with editable fields. 4. System accepts charger type update from AC to DC Fast Charger. 5. System saves changes to Firebase and displays "Changes saved successfully."	
Expected Result:	Station details are updated successfully and the changes are saved in the system.		
Actual Result:	Station charger type updated to DC Fast Charger. Changes saved and reflected in system for EV users.		
Status:	Pass		

Table 40:TC002-M6-B

Test Id:	TC002-M6-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Station without Authorization	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station ID of another owner	Priority:	High
Precondition:	Logged-in user not the owner of the selected station.		
Steps / Action		System Response	
1. The user attempts to access another owner’s station record. 2. The system displays the selected station details. 3. The user tries to open the edit station details page. 4. The user attempts to modify the station information. 1. 5. The user clicks on the “Save Changes” button.		1. Station owner (Owner B) dashboard loads. 2. System only shows stations owned by Owner B. Other stations are read-only. 3. Edit option not accessible for another owner's station. 4. Modification attempt blocked. 5. System displays: "You are not authorized to edit this station."	
Expected Result:	System displays an unauthorized access error and blocks the station update action.		
Actual Result:	Any user doesn’t have access to use app without registration or login.		
Status:	Pass		

Table 41:TC003-M6-A

Test Id:	TC003-M6-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Station Price Successfully	Test Case Executed by:	Zoya Khalid

Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Charging Price: 45 -> 50 PKR/unit	Priority:	High
Precondition:	The charging station is active and accessible to the station owner.		
Steps / Action		System Response	
1. The station owner navigates to the station pricing section. 2. The station owner selects the active charging station. 3. The station owner enters a new valid charging price. 4. The station owner updates the price value. 5. The station owner clicks on the “Save” button.		1. Station owner dashboard loads. 2. Station pricing section accessible from edit station page. 3. System accepts new price value: 50 PKR/kWh. 4. Price field updated. 5. System saves to Firebase. Updated price immediately visible to EV users in cost comparison.	
Expected Result:	Charging price is updated to 50 PKR/unit and immediately reflected for EV users.		
Actual Result:	Charging price updated to 50 PKR/kWh and reflected in system for EV users.		
Status:	Pass		

Table 42:TC003-M6-B

Test Id:	TC003-M6-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Station Price with Invalid Value	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Charging Price: -20	Priority:	High

Precondition:	The station owner is logged into the portal.	
Steps / Action		System Response
1. The station owner opens the station price update form. 2. The station owner enters an invalid negative price value. 3. The system validates the entered price input. 4. The station owner clicks on the “Save” button.		1. Price update form opens. 2. System accepts text input -20. 3. System validates: negative value detected, outside valid range. 4. System displays error: "Price must be a positive value." Save blocked.
Expected Result:	System displays a validation error and rejects the negative price value of -20.	
Actual Result:	System displayed validation error and rejected the negative price value of -20.	
Status:	Pass	

Table 43:TC004-M6-A

Test Id:	TC004-M6-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Mark Station as Unavailable (Maintenance)	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Availability: Available Unavailable	Priority:	High
Precondition:	1. The station ownership is verified. 2. The charging station is currently active		
Steps / Action		System Response	
1. The station owner navigates to the “Availability” section. 2. The station owner selects an active charging station.		1. Station owner dashboard loads. 2. Availability section shows current status: Open.	

3. The station owner changes the station status to “Unavailable”.	3. Status dropdown shows options: Open / Closed / Maintenance. User selects Maintenance.
Expected Result:	Station status is updated to Unavailable and the change is reflected in real time for EV users.
Actual Result:	Station status updated to Maintenance. Change reflected in real time on EV user map. Station shown as unavailable.
Status:	Pass

Table 44:TC004-M6-B

Test Id:	TC004-M6-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Status Without Internet Connection	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Status Change Attempt - Network Lost	Priority:	High
Precondition:	1. The station form is open. 2. The network connection is unavailable.		
Steps / Action		System Response	
1. The station owner opens the availability status page. 2. The station owner attempts to change the station status. 3. The network connection is interrupted during the process.		1. Availability page loads from cached data. 2. User attempts to change status. 3. Network connection lost. Firebase write operation fails. System displays: "No internet connection. Please try again."	
Expected Result:	System displays a network error message and the status change is not saved.		
Actual Result:	System displayed network error message. Status change not saved.		
Status:	Pass		

Table 45:TC005-M6-A

Test Id:	TC005-M6-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Operating Hours Correctly	Test Case Executed by:	Zoya Khalid
Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Opening: 08:00, Closing: 20:00	Priority:	Medium
Precondition:	The station owner is logged into the system.		
Steps / Action		System Response	
1. The station owner logs into the system. 2. The station owner opens the operating hours section. 3. The station owner updates the operating hours with valid time values. 4. The station owner saves the updated operating hours.		1. Station owner dashboard loads. 2. Operating hours section opens with time input fields. 3. System accepts 08:00 as valid opening time. 4. System accepts 20:00 as valid closing time. 5. System validates time range. Values are within 00:00-23:59. Saves to Firebase. Confirmation displayed.	
Expected Result:	Operating hours are updated to 08:00-20:00 and saved successfully in the system.		
Actual Result:	Operating hours updated to 08:00–20:00 and saved successfully.		
Status:	Pass		

Table 46:TC005-M6-B

Test Id:	TC005-M6-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Enter Invalid Time Format	Test Case Executed by:	Zoya Khalid

Module Name:	Charger Station Owner Portal	Test Case Execution Date:	07-07-2026 07-07-2026
Test Data:	Opening: 25:00, Closing: 30:00	Priority:	Medium
Precondition:	The charging station edit form is open and available for modification.		
Steps / Action		System Response	
1. The station owner opens the operating hours form. 2. The station owner enters invalid time values (e.g., 25:00, 30:00). 3. The station owner attempts to save the updated operating hours		Operating hours form opens. System accepts text input 25:00 and 30:00. System validates time values. 25:00 and 30:00 exceed valid 24-hour range. Error displayed: "Invalid time format."	
Expected Result:	System shows "Invalid time format" error.		
Actual Result:	Error shown.		
Status:	Pass		

Table 47:Test Case-7(Module 07)

Test Id:	TC-07	Test Case Designed by:	Sarwat Naveed
Test Case Title:	EV Owner Portal profile management works correctly	Test Case Executed by:	Zoya Khalid
Module Name:	EV Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Name, email, password, profile information, saved vehicle profile	Priority:	High
Precondition:	EV Owner is logged in to the application.		
Steps /Action		System Response	
1. Open EV Charging Application 2. Login as EV Owner 3. Open EV Owner Portal 4. View profile information 5. Update profile information 6. Access saved vehicle profiles		1. Application opens and login screen appears. EV Owner credentials verified and dashboard loads. 2. EV Owner Portal accessible from dashboard. 3. Profile page loads with current information. Profile edit fields become active.	

	4. Saved vehicle profiles listed in vehicle management section.
Expected Result:	System should allow EV Owner to view and update profile information and access saved vehicle profiles successfully.
Actual Result:	EV Owner portal worked correctly. Profile view, profile update, and vehicle profile access all functioned as expected.
Status:	Pass

Table 48:TC001-M7-A

Test Id:	TC001-M7-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	View Profile Information Successfully	Test Case Executed by:	Zoya Khalid
Module Name:	EV Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	User: Sara Ahmed	Priority:	High
Precondition:	EV owner is logged into the system.		
Steps / Action		System Response	
1. The user logs into the EV Owner Portal using valid credentials. 2. The user navigates to the “My Profile” section. 3. The user opens the profile page. 4. The user views the displayed personal information. 5. The user verifies that the profile details are shown correctly.		1. Login screen authenticates Sara Ahmed's credentials. 2. EV Owner dashboard loads. My Profile option visible in navigation. 3. Profile page opens. 4. System fetches user data from Firebase and displays name, email, and role. 5. All profile fields display correct and complete information.	
Expected Result:	Profile page displays all personal information of Sara Ahmed correctly.		

Actual Result:	Profile page displayed Sara Ahmed's name, email, and role correctly.
Status:	Pass

Table 49:TC001-M7-B

Test Id:	TC001-M7-B	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Attempt View Without Login	Test Case Executed by:	Zoya Khalid
Module Name:	EV Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	None	Priority:	High
Precondition:	EV owner is logged into the system.		
Steps / Action		System Response	
1. The user attempts to access the “My Profile” page directly. 2. The user is not logged into the system. 3. The user tries to open the profile section without authentication. 4. The user is redirected away from the profile page. 5. The user is taken to the login page instead of viewing profile details.		1. Direct access to profile page attempted with no active session. 2. System detects no authenticated user session. 3. System blocks profile page access and redirects to login page automatically. 4. Login page displayed instead of profile. 5. User cannot proceed without valid authentication.	
Expected Result:	System denies access and redirects the unauthenticated user to the login page.		
Actual Result:	System denied access and redirected unauthenticated user to login page.		
Status:	Pass		

Table 50:TC002-M7-A

Test Id:	TC002-M7-A	Test Case Designed by:	Sarwat Naveed
Test Case Title:	Update Profile Successfully	Test Case Executed by:	Zoya Khalid
Module Name:	EV Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Name: Sara Ahmed, Sara Ameen, Email: sara@gmail.com	Priority:	High
Precondition:	1. The user is logged into the EV Owner Portal with valid credentials. 2. The user has access to the profile edit screen.		
Steps / Action		System Response	
1. The user logs into the EV Owner Portal using valid credentials. 2. The user navigates to the profile section. 3. The user opens the profile edit screen. 4. The user updates the name and email fields with valid information. 5. The user clicks on the “Save” button.		1. Login successful. Dashboard loads. 2. Profile section opened from navigation menu. 3. Profile edit screen opened. Editable fields highlighted. 4. System accepts updated name: Sara Ameen and valid email. 5. System validates inputs. Saves to Firebase. Displays: "Profile updated successfully."	
Expected Result:	Profile is updated successfully and the new name and email are saved in the system.		
Actual Result:	Profile updated successfully. New name Sara Ameen saved and displayed in system.		
Status:	Pass		

Table 51:TC002-M7-B

Test Id:	TC002-M7-B	Test Case Designed by:	Sarwat Naveed
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Test Case Title:	Update Profile with Invalid Email Format	Test Case Executed by:	Zoya Khalid
Module Name:	EV Owner Portal	Test Case Execution Date:	07-07-2026
Test Data:	Email: sara@	Priority:	High
Precondition:	The user is logged in as an authorized EV owner.		
Steps / Action		System Response	
1. The user logs into the EV Owner Portal using valid credentials. 2. The user navigates to the profile section. 3. The user opens the profile edit screen. 4. The user enters an invalid email format in the email field. 5. The user clicks on the “Save” button.		1. Login successful. Dashboard loads. 2. Profile section opened. 3. Edit screen opened. 4. System detects invalid email format: sara@ (missing domain). 5. System displays: "Please enter a valid email address." Changes not saved.	
Expected Result:	System displays an invalid email format error and does not save the profile changes.		
Actual Result:	System displayed invalid email format error and did not save the profile changes.		
Status:	Pass		

Table 52:Test Case-8(Module 08)

Test Id:	TC-M8	Test Case Designed by:	Waleed
Test Case Title:	Admin Management Portal	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	1. Admin Email: admin@evsystem.com 2. Password: Admin@123 3. Station ID: CS-1003 4. User Name: Ali Khan 5. Performance Data: Last 24 Hours 6. Pending Station: CS-1004	Priority:	High
Precondition:		1. Administrator account already exists. 2. Administrator is logged into the system. 3. The Admin Management Portal is accessible. 4. At least one charging station and one user account already exist	
Steps / Action		System Response	
1. Open the EV Charging Application. 2. Login using valid administrator credentials. 3. Open the Admin Management Portal. 4. Add a new charging station. 5. Update the charging station information. 6. Delete an existing charging station. 7. View the System Performance Dashboard. 8. Create a new user account. 9. Approve a pending charging station request. 10. Logout from the system.		1. The administrator is authenticated successfully. 2. The Admin Dashboard is displayed. 3. The new charging station is added successfully. 4. Updated station information is saved correctly. 5. The selected charging station is deleted successfully. 6. System performance statistics are displayed correctly.	

	7. A new user account is created successfully. 8. The pending charging station status changes to Approved. 9. A confirmation message is displayed after each successful operation. 10. The administrator is logged out successfully.
Expected Result:	All administrative functions should execute successfully, and every operation should update the database correctly without any validation or system errors.
Actual Result:	All administrative operations were completed successfully. The system processed each request correctly, updated the database accurately, and displayed the expected confirmation messages.
Status:	Pass

Table 53 :TC001-M8-A

Test Id:	TC001-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Admin Login with Valid Credentials	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Email: admin@evsystem.com Password: Admin@123	Priority:	High
Precondition:		Admin account is already registered in the system. Admin is not currently logged in.	

Steps / Action	System Response
11. Navigate to the Admin Login page. 12. Enter a valid admin email address. 13. Enter the correct password. 14. Click the Login button. 15. Wait while the system verifies the credentials. 16. Access the Admin Dashboard.	11. The system validates the entered email and password 12. The system authenticates the admin account. 13. The system creates a secure login session. 14. The system loads the Admin Dashboard. 15. The dashboard displays users, charging stations, vehicles, and system statistics.
Expected Result:	The admin should be successfully authenticated and redirected to the Admin Dashboard .
Actual Result:	The admin was successfully authenticated, and the Admin Dashboard loaded correctly.
Status:	Pass

Table 54:TC001-M8-B

Test Id:	TC001-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Admin Login with Invalid Credentials	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Email:admin@evsystem.com Password:Wrong@123	Priority:	High
Precondition:	Admin login page is open. No active admin session exists		
Steps / Action	System Response		
1. Open the Admin Login page. 2. Enter an invalid email or password. 3. Click the Login button.	1. The system validates the entered credentials. 2. Authentication fails.		

4. Wait for authentication.	3. The system displays "Invalid Email or Password." 4. The login request is rejected. 5. The admin remains on the Login page
Expected Result:	The system should reject the login request and display an appropriate error message without granting access.
Actual Result:	The login request was rejected successfully, and an error message was displayed while access to the Admin Dashboard was denied.
Status:	Pass

Table 55:TC002-M8-A

Test Id:	TC002-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Update Charging Station Successfully	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station ID: CS-1001; Updated Capacity: 6 ports	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal.	
Steps / Action		System Response	
1. Login as Admin. 2. Open Manage Stations. 3. Select an existing charging station. 4. Click Edit. 5. Update charger type, charging cost or availability. 6. Click Update.		1. The system validates the updated information. 2. The existing station record is modified. 3. The updated data is saved in the database. 4. A success message is displayed. 5. The updated station information appears in the station list.	

Expected Result:	The selected charging station information should be updated successfully and reflected in the station list.
Actual Result:	The station details were updated successfully, and the latest information was displayed in the station management panel.
Status:	Pass

Table 56:TC002-M8-B

Test Id:	TC002-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Update Station with Invalid Data	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station ID: CS-9999	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal.	
Steps / Action		System Response	
1. Login as Admin. 2. Open the station edit page. 3. Enter invalid information (empty station name or negative charging cost). 4. Click Update.		1. The system validates the entered information. 2. Invalid fields are detected. 3. Appropriate validation messages are displayed. 4. Changes are rejected. 5. Existing station information remains unchanged.	

Expected Result:	The system should reject invalid information and keep the existing station details unchanged.
Actual Result:	The invalid data was rejected successfully, and the previous station information remained unchanged.
Status:	Pass

Table 57:TC003-M8-A

Test Id:	TC003-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Delete Charging Station	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station Name: EV Fast Charge Lahore Action: Delete	Priority:	High
Precondition:	- Admin is logged in. - At least one charging station exists. - Station Management page is open.		
Steps / Action		System Response	
- Login as Admin. - Open Manage Stations. - Select a charging station. - Click Delete. - Confirm the deletion.		- The system asks for confirmation. - The selected station is deleted from the database. - The station list is refreshed. - A success notification is displayed. - The deleted station no longer appears in the list.	
Expected Result:		The selected charging station should be permanently removed from the database and no longer appear in the station list.	

Actual Result:	The charging station was deleted successfully, and the updated station list no longer displayed the deleted record.
Status:	Pass

Table 58:TC003-M8-B

Test Id:	TC003-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Cancel Delete Operation	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station Name: EV Fast Charge Lahore Confirmation: Cancel	Priority:	High
Precondition:		Admin is logged in. Existing charging station is selected. Delete confirmation dialog is displayed.	
Steps / Action		System Response	
1. Login as Admin. 2. Select a charging station. 3. Click Delete. 4. Select Cancel in the confirmation dialog.		1. The delete request is cancelled. 2. No record is removed. 3. The station remains available in the list. 4. The system returns to the Station Management page.	
Expected Result:		The delete operation should be cancelled, and the selected charging station should remain unchanged.	
Actual Result:		The delete request was cancelled successfully, and the charging station remained available in the station list.	

Status:	Pass
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Table 59:TC004-M8-A

Test Id:	TC004-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Manage User Account	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle ID: EV-2001; Model: Tesla Model 3; Owner: John Doe	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal.	
Steps / Action		System Response	
1. Login as Admin. 2. Open Manage Users. 3. Select a user account. 4. Edit user status or account details. 5. Click Save Changes.		1. The system validates the updated information. 2. User account details are updated. 3. Changes are stored in the database. 4. A success message is displayed. 5. The updated user information appears in the user list.	
Expected Result:		The selected user account should be updated successfully, and the changes should be reflected in the user management section.	
Actual Result:		The user account was updated successfully, and the modified information was displayed correctly in the management panel	
Status:		Pass	

Table 60:TC004-M8-B

Test Id:	TC004-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Unauthorized User Access	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Vehicle ID: EV-2001; Invalid VIN format entered	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal.	
Steps / Action		System Response	
1. Login as a normal EV User. 2. Attempt to open the Admin Dashboard using the admin URL or restricted menu. 3. Wait for authorization.		1. The system verifies the user's role. 2. The system detects insufficient privileges. 3. Access to the Admin Dashboard is denied. 4. An "Access Denied" or "Unauthorized Access" message is displayed. 5. The user is redirected to the appropriate page or login screen.	
Expected Result:		The system should deny access to unauthorized users and display an appropriate access restriction message.	
Actual Result:		The unauthorized user was prevented from accessing the Admin Dashboard, and an access denied message was displayed..	
Status:		Pass	

Table 61:TC005-M8-A

Test Id:	TC005-M8-A	Test Case Designed by:	Waleed
Test Case Title:	View System Performance Dashboard	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Performance metrics for last 24 hours	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal. Performance data has been recorded for the last 24 hours.	
Steps / Action		System Response	
1. Open the EV Charging Application. 2. Login as Administrator. 3. Open the Admin Management Portal. 4. Click System Performance Dashboard. 5. Select Last 24 Hours and click View Report.		1. The administrator is authenticated successfully. 2. The dashboard loads successfully. 3. The system retrieves CPU usage, memory usage, active users, and server status. 4. Performance statistics are displayed in charts and summary cards. 5. The requested dashboard is displayed successfully.	
Expected Result:		The system should display the latest performance statistics and system health information accurately.	
Actual Result:		The dashboard loaded successfully, and all performance statistics were displayed correctly.	
Status:		Pass	

Table 62:TC005-M8-B

Test Id:	TC005-M8-B	Test Case Designed by:	Waleed
Test Case Title:	System Performance Data Unavailable	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	No performance data available for selected period	Priority:	High
Precondition:		Administrator is logged into the system. No performance records exist for the selected time period.	
Steps / Action		System Response	
1. Open the EV Charging Application. 2. Login as Administrator. 3. Open System Performance Dashboard. 4. Select a time period with no available records. 5. Click View Report.		1. The system checks the selected period. 2. No performance records are found. 3. The system displays "No performance data available for the selected period." 4. No dashboard statistics are displayed.	
Expected Result:		The system should inform the administrator that no performance data is available without generating an error.	
Actual Result:		The appropriate information message was displayed, and no incorrect data appeared.	
Status:		Pass	

Table 63:TC006-M8-A

Test Id:	TC006-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Create User Account	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Username: admin_user01; Email: user01@example.com; Role: Admin	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal.	
Steps / Action		System Response	
1. Open the Admin Management Portal. 2. Navigate to User Management. 3. Click Create User. 4. Enter username, email and role. 5. Click Save		1. The system validates all entered information. 2. The system checks whether the email already exists. 3. A new user account is created. 4. The user is added to the user list. 5. A success message is displayed.	
Expected Result:		The new user account should be created successfully and displayed in the user management list.	
Actual Result:		The user account was created successfully and appeared in the user list.	
Status:		Pass	

Table 64:TC006-M8-B

Test Id:	TC006-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Delete User Account	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Username: admin_user01	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal. The selected user account already exists.	
Steps / Action		System Response	
1. Open User Management. 2. Select the required user account. 3. Click Delete User. 4. Confirm the deletion. 5. Refresh the user list.		1. The system requests confirmation. 2. The selected user account is deleted. 3. The database is updated. 4. The user list is refreshed. 5. A success message is displayed.	
Expected Result:		The selected user account should be removed successfully from the system.	
Actual Result:		The selected user account should be removed successfully from the system.	
Status:		Pass	

Table 65:TC007-M8-A

Test Id:	TC007-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Approve Newly Added Station	Test Case Executed by:	Asad

Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station ID: CS-1003 (pending approval)	Priority:	High
Precondition:		Administrator is logged in and has access to the Admin Management Portal. A charging station is available with Pending Approval status.	
Steps / Action		System Response	
1. Open the Admin Management Portal. 2. Navigate to Pending Station Requests. 3. Select Station ID CS-1003. 4. Review the station information. 5. Click Approve.		1. The system validates the station information. 2. The station status changes from Pending to Approved. 3. The database is updated. 4. The station becomes available for users. 5. A confirmation message is displayed.	
Expected Result:		The charging station should be approved successfully and become visible to EV users.	
Actual Result:		The station was approved successfully and became available in the system.	
Status:		Pass	

Table 66:TC007-M8-B

Test Id:	TC007-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Reject Invalid Station Record	Test Case Executed by:	Asad

Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station ID: CS-1004 (incomplete/invalid data)	Priority:	High
Precondition:		Administrator is logged in. A charging station contains incomplete or invalid information.	
Steps / Action		System Response	
1. Open Pending Station Requests. 2. Select Station ID CS-1004. 3. Review the station details. 4. Click Reject. 5. Confirm the rejection.		1. The system verifies the station information. 2. The invalid station request is rejected. 3. The station remains unavailable to users. 4. The database status is updated. 5. A rejection confirmation message is displayed.	
Expected Result:		The invalid station record should be rejected and should not appear in the approved station list.	
Actual Result:		The invalid station record was rejected successfully and remained unavailable.	
Status:		Pass	

Table 67:TC008-M8-A

Test Id:	TC008-M8-A	Test Case Designed by:	Waleed
Test Case Title:	Verify Approved Station Visibility	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026

Test Data:	Station ID: CS-1003 (approved)	Priority:	High
Precondition:		Administrator has approved Station ID CS-1003. The station exists in the database.	
Steps / Action		System Response	
1. Open the Admin Management Portal. 2. Navigate to Approved Stations. 3. Search for Station ID CS-1003. 4. Open the station details. 5. Verify its approval status.		1. The system retrieves the station record. 2. The approved status is displayed. 3. The station appears in the approved station list. 4. Station information is displayed correctly. 5. No validation errors occur.	
Expected Result:		The approved charging station should be visible in the approved station list with the correct status.	
Actual Result:		The approved charging station appeared successfully with the correct approval status.	
Status:		Pass	

Table 68:TC008-M8-B

Test Id:	TC008-M8-B	Test Case Designed by:	Waleed
Test Case Title:	Prevent Duplicate Station Approval	Test Case Executed by:	Asad
Module Name:	Admin Management Portal	Test Case Execution Date:	07-07-2026
Test Data:	Station ID: CS-1003 (already approved)	Priority:	High

Precondition:	Administrator is logged in. Station ID CS-1003 has already been approved.
Steps / Action	System Response
1. Open Approved Stations. 2. Select Station ID CS-1003. 3. Click Approve again. 4. Confirm the action if prompted.	1. The system checks the station status. 2. It detects that the station is already approved. 3. Duplicate approval is prevented. 4. A warning message is displayed. 5. No duplicate record is created
Expected Result:	The system should prevent duplicate approval and notify the administrator that the station is already approved.
Actual Result:	The duplicate approval request was blocked successfully, and a warning message was displayed.
Status:	Pass

Table 69 Test Case-9 (Module 09)

Test Id:	TC-09	Test Case Designed by:	Muhammad Asad
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Test Case Title:	Rating and Review with ML Integration works correctly	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Rating: 5, Review: “Charging was fast and staff was helpful.”	Priority:	High
Precondition:		EV Owner is logged in and selected charging station details are available.	
Steps /Action		System Response	
1. Open EV Charging Application 2. Login as EV Owner 3. Open a charging station detail page 4. Submit star rating and written review 5. Run ML sentiment/spam analysis 6. View updated station rating and review summary		1. Application opens and EV Owner logs in. Station detail page opens with rating and review section. 2. Star rating and text fields available. Review text sent to FastAPI /analyze_review. DistilBERT returns sentiment label and confidence score. 3. Result stored in Firebase. 4. Updated station rating and approved reviews displayed.	
Expected Result:		System should accept valid ratings and reviews, analyze review sentiment, flag spam or inappropriate content, and display updated station feedback.	
Actual Result:		Rating and review system functioned correctly with ML integration. Sentiment analysis returned correct labels. Admin moderation panel showed flagged reviews.	
Status:		Pass	

Table 70 :TC001-M9-A

Test Id:	TC001-M9-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Submit valid station rating	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Rating: 4 stars	Priority:	Medium
Precondition:		EV Owner is logged in and a verified charging station is selected.	
Steps /Action		System Response	
1. Open station detail page 2. Select rating value between 1 and 5 3. Click “Submit Rating”		1. Station detail page opens. 2. Star rating selector displays 1-5 stars. User selects 4 stars. 3. Submit Rating clicked. System saves rating to Firestore against station ID.	

	Average station rating recalculated and updated.
Expected Result:	Rating should be submitted successfully and saved against the selected charging station.
Actual Result:	4 -star rating submitted and saved successfully. Station average rating updated in real time.
Status:	Pass

Table 71 TC001-M9-B

Test Id:	TC001-M9-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	Submit invalid station rating	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Rating: 0 or 6 stars	Priority:	Medium
Precondition:	EV Owner is logged in and station detail page is open.		
Steps /Action	System Response		
1. Open station detail page 2. Enter or select rating outside allowed range 3. Click "Submit Rating"	1. Station detail page opens. 2. Star rating UI restricts selection to 1-5 only. Values outside this range not selectable. 3. Submit attempted with out-of-range value. System displays: "Rating must be between 1 and 5."		
Expected Result:	System should reject the rating and display validation message that rating must be between 1 and 5.		
Actual Result:	System prevented selection of rating outside 1–5 range. Validation error displayed.		
Status:	Pass		

Table 72 TC002-M9-A

Test Id:	TC002-M9-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Submit written review with acceptable content	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Review: "Station was clean and charger worked properly."	Priority:	Medium
Precondition:	EV Owner is logged in and station rating page is available.		

Steps /Action	System Response
1. Open station detail page 2. Enter meaningful written review 3. Click “Submit Review”	1. Station detail page opens. 2. Review text field accepts input. 3. Submit Review clicked. System sends review text to FastAPI POST /analyze_review endpoint. DistilBERT tokenizes text and returns: Positive, confidence: 0.92. Result stored in Firestore with review text and star rating.
Expected Result:	Review should be accepted, saved, and sent to ML model for sentiment and spam checking.
Actual Result:	Review accepted and saved. Sentiment label Positive with confidence 0.92 stored in Firestore.
Status:	Pass

Table 73 TC002-M9-B

Test Id:	TC002-M9-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	Submit empty written review	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Review text: Empty	Priority:	Medium
Precondition:	EV Owner is logged in and review form is open.		
Steps /Action	System Response		
1. Open review form 2. Leave review text field empty 3. Click “Submit Review”	1. Review form opened. 2. Review text field left empty. 3. Submit clicked. System detects empty field. Displays: "Please enter your review before submitting."		
Expected Result:	System should display a validation message asking the user to enter review text.		
Actual Result:	System displayed validation message. Review not submitted.		
Status:	Pass		

Table 74 TC003-M9-A

Test Id:	TC003-M9-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Display overall station rating and user reviews	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Station has multiple approved ratings and reviews	Priority:	High
Precondition:		Station has existing approved user feedback.	
Steps /Action		System Response	
1. Open station detail page 2. Scroll to ratings and reviews section 3. View overall rating, comments, and feedback summary		1. Station detail page opens. 2. Reviews section loaded from Firestore. 3. System displays: overall average star rating, individual user reviews with reviewer name, date, written text, and sentiment label. Review count also displayed.	
Expected Result:		System should display calculated overall station rating, user comments, and feedback summary correctly.	
Actual Result:		Overall station rating and all approved user reviews displayed correctly with sentiment labels.	
Status:		Pass	

Table 75 TC003-M9-B

Test Id:	TC003-M9-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	Display review section when no reviews exist	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Station with zero reviews	Priority:	Medium
Precondition:		Selected station has no existing user reviews.	
Steps /Action		System Response	
1. Open station detail page 2. View ratings and reviews section		1. Station detail page opens. 2. Reviews section loaded. Firestore returns empty list for this station. System displays: "No reviews available yet."	
Expected Result:		System should display a clear message such as "No reviews available yet" instead of showing blank or incorrect data.	
Actual Result:		System displayed "No reviews available yet" message instead of blank section.	

Status:	Pass
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Table 76 TC004-M9-A

Test Id:	TC004-M9-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Generate ML sentiment score for review	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Review: "The charger was working well and service was good."	Priority:	High
Precondition:		ML review analysis service/API is available.	
Steps /Action		System Response	
1. Submit a valid written review 2. Send review text to ML sentiment model 3. Process model output 4. Store sentiment result with review		1. Valid written review submitted. 2. System sends POST request to FastAPI: { "review_text": "The charger was working well and service was good.", "star_rating": 4 } 3. DistilBERT model tokenizes text and runs inference. 4. System receives: { "sentiment": "Positive", "confidence": 0.91 } 5. Sentiment label and confidence score stored in Firestore with review record.	
Expected Result:		System should generate and store sentiment score/class as Positive, Neutral, or Negative.	
Actual Result:		ML model returned Positive sentiment with confidence 0.91. Stored correctly in Firestore.	
Status:		Pass	

Table 77 TC004-M9-B

Test Id:	TC004-M9-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	ML detects spam, fake, or inappropriate review	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Rating: 5, Review: repeated words/offensive text/inconsistent sentiment	Priority:	High
Precondition:		ML spam/inappropriate content detection is enabled.	
Steps /Action		System Response	

1. Submit a suspicious or inappropriate review 2. Send review to ML filtering service 3. Compare rating and sentiment consistency 4. Check toxic/spam probability	1. Suspicious review submitted (5-star rating but text: "worst place ever broken charger avoid"). 2. System sends review to FastAPI. DistilBERT detects Negative sentiment – mismatch with 5-star rating. 3. Mismatch flag raised. Spam/toxicity probability calculated. 4. flag_label set to "Flagged" in Firestore. moderation_label set to "mismatch". 5. Review not displayed publicly. Appears in admin audit panel for review.
Expected Result:	System should flag or reject spam, fake, toxic, or inappropriate review and prevent unreliable feedback from being publicly displayed.
Actual Result:	ML model detected rating-sentiment mismatch. Review flagged and visible in admin audit panel only. Not displayed to public.
Status:	Pass

Table 78 TC005-M9-A

Test Id:	TC005-M9-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Admin removes inappropriate review	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	Flagged Review ID: REV-102	Priority:	Medium
Precondition:	Admin is logged in and a flagged review exists.		
Steps /Action	System Response		
1. Login as Admin 2. Open review monitoring section 3. Select flagged/inappropriate review 4. Click “Remove Review” 5. Confirm removal	1. Admin audit page opens with flagged reviews list. 2. Review monitoring section shows all ML-flagged reviews with flag reason. 3. Review REV-102 selected. Review content and flag reason visible. 4. Remove Review clicked. Confirmation dialog: "Remove this review permanently?" 5. Confirmed. System deletes review from Firestore. Station review list updated. Average rating recalculated. Audit entry created.		

Expected Result:	System should remove the inappropriate review and update station review list accordingly.
Actual Result:	Review REV-102 removed from station. Review list and rating updated. Audit log entry recorded with admin ID and timestamp.
Status:	Pass

Table 79 TC005-M9-B

Test Id:	TC005-M9-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	Non-admin attempts to remove review	Test Case Executed by:	Zoya Khalid
Module Name:	Rating and Review with ML Integration	Test Case Execution Date:	07-07-2026
Test Data:	EV Owner tries to delete another user's review	Priority:	Medium
Precondition:	EV Owner is logged in and review list is visible.		
Steps /Action	System Response		
1. Login as EV Owner 2. Open station review section 3. Attempt to remove another user's review	1. EV Owner dashboard loads. 2. Station review section shows list of reviews. 3. No remove/delete button available for any review. Role-based access control enforced. EV Owner role cannot delete reviews.		
Expected Result:	System should block the action and allow only Admin to remove inappropriate or misleading reviews.		
Actual Result:	Remove option not available for EV Owner. Action blocked by role-based access control.		
Status:	Pass		

Table 80 Test Case-10 (Module 10)

Test Id:	TC-10	Test Case Designed by:	Muhammad Asad
Test Case Title:	Amenities module works correctly	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	Amenities: Wi-Fi, Restroom, Café, Parking, Rest Area, Shopping	Priority:	Medium

Precondition:	Charging station data is available and user/station owner has valid access according to role.
Steps /Action	System Response
1. Open EV Charging Application 2. Open station details 3. View available amenities 4. Login as station owner and update amenities 5. Filter stations based on required amenities	1. Application opens. 2. Station detail page loads. 3. Amenities section visible with listed items. Station owner dashboard shows amenities checkboxes in edit station page. 4. System saves updated amenities to Firebase. 5. Updated amenities visible on station detail page for EV users.
Expected Result:	System should display station amenities, allow station owner to update amenities, and filter charging stations according to selected amenities.
Actual Result:	Amenities module worked correctly. Display and update functions performed as expected.
Status:	Pass

Table 81 TC001-M10-A

Test Id:	TC001-M10-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	View available amenities for selected station	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	Amenities: Wi-Fi, Restroom, Café, Parking	Priority:	Medium
Precondition:	EV Owner is logged in and selected station has amenities information.		
Steps /Action	System Response		
1. Open charging station locator 2. Select a charging station 3. Open station detail page 4. View amenities section	1. Charging station locator map opens. 2. Station selected on map. Detail bottom sheet or page opens. 3. Station detail page fully loaded. 4. Amenities section displays: Wi-Fi, Restroom, Café, Parking icons/labels from Firestore data.		
Expected Result:	System should display all amenities linked to the selected charging station.		
Actual Result:	All amenities (Wi-Fi, Restroom, Café, Parking) displayed correctly on station detail page.		

Status:	Pass
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Table 82 TC001-M10-B

Test Id:	TC001-M10-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	View station when amenities are not available	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	Station record with no amenities selected	Priority:	Medium
Precondition:		EV Owner is logged in and selected station has no amenities data.	
Steps /Action		System Response	
1. Open station detail page 2. View amenities section		1. Station detail page opens. 2. Amenities section loaded. Firebase returns empty amenities list for this station. System displays: "No amenities listed."	
Expected Result:		System should show a clear message such as “No amenities listed” and should not display incorrect amenities.	
Actual Result:		System displayed "No amenities listed" message correctly instead of blank section.	
Status:		Pass	

Table 83 TC002-M10-A

Test Id:	TC002-M10-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Station owner updates amenities information	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	Add: Café, Restroom, Parking	Priority:	Medium
Precondition:	Registered station owner is logged in and owns the selected station.		
Steps /Action	System Response		
1. Login as station owner 2. Open owner dashboard 3. Select own charging station 4. Update amenities list 5. Save changes		1. Station owner dashboard loads. 2. My Stations list opens. 3. Own charging station selected. Edit page opens.	

	4. Amenities checkboxes section shows available options. Café, Restroom, Parking checkboxes selected. 5. Save Changes clicked. System updates amenities list in Firebase. Confirmation displayed. Updated amenities visible on station detail page for EV users.
Expected Result:	System should save updated amenities and display them on the station detail page.
Actual Result:	Amenities updated and saved to Firestore. Café, Restroom, Parking visible to EV users on station detail page.
Status:	Pass

Table 84 TC002-M10-B

Test Id:	TC002-M10-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	Unauthorized user attempts to update amenities	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	EV Owner or different station owner attempts update	Priority:	High
Precondition:	User is logged in but does not own the selected station.		
Steps /Action	System Response		
1. Login with unauthorized account 2. Open station amenities update option 3. Attempt to change amenities 4. Click save	1. EV Owner dashboard loads after login. 2. EV Owner role has no access to station edit page. Amenities update option not visible in EV Owner interface. 3. Direct access attempt blocked. Permission error displayed. 4. Save operation never reached. Access denied.		
Expected Result:	System should deny the update and display permission error because only registered station owner can update amenities for own station.		
Actual Result:	System denied access. EV Owner cannot update amenities. Permission error displayed.		
Status:	Pass		

Table 85 TC003-M10-A

Test Id:	TC003-M10-A	Test Case Designed by:	Muhammad Asad
Test Case Title:	Filter stations by selected amenities	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	Selected filter: Restroom and Parking	Priority:	Medium
Precondition:		Station locator contains stations with different amenities.	
Steps /Action		System Response	
1. Open charging station locator 2. Open filter options 3. Select required amenities 4. Apply filter		1. Charging station locator opens. 2. Filter options panel opened. 3. Restroom and Parking checkboxes selected. 4. Filter applied. System attempts to filter station list by selected amenities.	
Expected Result:		System should display only charging stations that match the selected amenities.	
Actual Result:		System display selected amenities.	
Status:		Pass	

Table 86 TC003-M10-B

Test Id:	TC003-M10-B	Test Case Designed by:	Muhammad Asad
Test Case Title:	No station matches selected amenities filter	Test Case Executed by:	Zoya Khalid
Module Name:	Amenities	Test Case Execution Date:	07-07-2026
Test Data:	Selected filter: Café + Shopping + Rest Area, no matching station	Priority:	Medium
Precondition:		No station in current area matches all selected amenities.	
Steps /Action		System Response	
1. Open amenities filter 2. Select amenities combination with no match 3. Apply filter		1. Amenities filter panel opens. 2. Café, Shopping, Rest Area selected as filter. 3. Filter applied. System searches Firestore for matching stations. No station has all three selected amenities. 4. System displays: "No stations match your selected amenities. Try adjusting your filters."	
Expected Result:		System should display a no-results message and allow the user to clear or change filters.	
Actual Result:		No matching stations found. System displayed no-results message with option to clear filters.	
Status:		Pass	

2.14.1 Unit Testing (Test Cases)

Unit Testing 1: Credential Validation Function

Testing Objective: To ensure that the system correctly validates user credentials (email & password) before authentication.

Table 87: Test Case-M1 (Unit 1)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Validate correct email and password format	Email: user@gmail.com Password: user@1234	Input accepted as valid	Pass
2.	Validate invalid email format	Email: usertest.com	Validation error displayed	Pass
3.	Validate empty email field	Email: —	Error “Email required”	Pass
4.	Validate empty password field	Password: —	Error “Password required”	Pass
5.	Validate password length	Password: 1234	Error “Password too short” or “Weak password” displayed	Pass

Unit Testing 2: Authentication Function

Testing Objective: To ensure that the system correctly authenticates registered users and blocks invalid access attempts.

Table 88: Test Case-M1 (Unit 2)

No.	Test case/Test script	Attribute and value	Expected result	Result
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1.	Login with valid registered credentials	Email: user@gmail.com Password: user@1234	User authenticated successfully	Pass
2.	Login with incorrect password	Email: user@gmail.com Password: wrong123	Error “Invalid credentials” displayed	Pass
3.	Login with unregistered email	Email: notfound@gmail.com	Access denied and error displayed	Pass
4.	Access protected screen without login	No active session	System restricts access and redirects to login	Pass
5.	Logout after valid login	Active session	User session terminated successfully	Pass

Unit Testing 3: Password Recovery Function

Testing Objective: To ensure that the system provides secure password recovery for registered users.

Table 89: Test Case-M1 (Unit 3)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Recover password with registered email	Email: user@gmail.com	Password recovery process initiated successfully	
2.	Recover password with unregistered email	Email: abc@gmail.com	Error “Email not found” displayed	
3.	Submit empty email in password recovery	Email: —	Error “Email required” displayed	
4.	Reset password with valid recovery flow	Registered account + valid recovery request	Password reset completed successfully	

Unit Testing 4: Encrypted Authentication / Security Check

Testing Objective: To ensure that authentication is managed securely and unauthorized access is prevented.

Table 90: Test Case-M1 (Unit 4)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Store user credentials securely during registration	Valid email and password	Credentials handled securely by the system	Pass
2.	Authenticate user through secure login flow	Valid registered credentials	Secure authentication completed successfully	Pass

3.	Reject unauthorized login attempt	Invalid credentials	Access denied	Pass
4.	Maintain access only for authenticated user	Active authenticated session	Authorized user can access system features	Pass

Unit Testing 5: GPS and Map Integration Function

Testing Objective: To ensure that the system correctly detects the user's location and displays nearby charging stations on the map.

Table 91: Test Case-M2 (Unit 5)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Detect user location with GPS enabled	GPS = ON	Current location detected successfully	Pass
2.	Display nearby stations on map	Valid current location	Nearby charging stations shown on map	Pass
3.	Access locator with GPS disabled	GPS = OFF	System prompts user to enable GPS/location service	Pass
4.	Load map view for charging stations	Internet available	Map loads successfully with station markers	Pass

Unit Testing 6: Station Details Display Function

Testing Objective: To ensure that the system displays the required charging station details correctly.

Table 92: Test Case-M2 (Unit 6)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	View selected station name	Station selected	Correct station name displayed	Pass
2.	View selected station charger type	Charger type available	Correct charger type displayed	Pass
3.	View selected station cost per unit	Cost data available	Correct cost per unit displayed	Pass
4.	View selected station status	Status available	Current station status displayed correctly	Pass

Unit Testing 7: Station Filter Function

Testing Objective: To ensure that users can filter charging stations by location, charger compatibility, and price.

Table 93: Test Case-M2 (Unit 7)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Filter stations by location	Location: Lahore	Only stations in Lahore displayed	Pass
2.	Filter stations by charger compatibility	Charger Type: Fast Charger	Only compatible stations displayed	Pass
3.	Filter stations by price	Price Range: PKR 80–100	Only stations within selected price range displayed	Pass
4.	Apply multiple filters together	Location + charger type + price	Matching stations displayed correctly	Pass

Unit Testing 8: Battery Percentage Input Validation

Testing Objective: To ensure that the user can manually enter battery percentage correctly and invalid values are rejected.

Table 94: Test Case-M3 (Unit 8)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Enter valid battery percentage	Battery: 35%	Input accepted successfully	Pass
2.	Enter battery percentage above valid range	Battery: 120%	Validation error displayed because user have only a slide	Pass

			bar to adjust battery percentage	
3.	Enter battery percentage below valid range	Battery: -5%	There is no range below 0 because user was given a slide bar to adjust battery	Pass

Unit Testing 9: Vehicle Data and Travel Distance Calculation

Testing Objective: To ensure that the system correctly uses selected vehicle details to calculate remaining travel distance.

Table 95: Test Case-M3 (Unit 9)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Retrieve selected vehicle details	Vehicle selected from Vehicle Management module	Vehicle model, battery capacity, and range loaded successfully	Pass
2.	Calculate remaining travel distance	Battery: 35%, Range: 320 km	Remaining travel distance calculated correctly	Pass
3.	Start calculation without selected vehicle	No selected vehicle	System prompts user to select vehicle first	Pass

Unit Testing 10: Nearby Charging Station Suggestion Function

Testing Objective: To ensure that the system suggests nearest charging stations within the calculated travel range.

Table 96: Test Case-M3 (Unit 10)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Suggest nearby charging stations within travel range	Valid battery % + selected vehicle	Nearest charging stations displayed successfully	Pass
2.	Display route suggestions using GPS/map integration	GPS enabled	Reachable stations shown correctly on map	Pass
3.	No station available within range	Low battery + long route	Alert that no station available	Pass

Unit Testing 11: Vehicle Name Validation

Testing Objective: To ensure that vehicle is added successfully with valid input

Table 97:Test Case-M4 (Unit 11)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Enter Valid Vehicle name	Name: Tesla	System accepts input	Pass
2.	Enter empty vehicle name	Name: (empty)	System shows error message and does not accept input	Pass

Unit Testing 12: Empty Field Validation

Testing Objective: To ensure required fields are not empty

Table 98:Test Case-M4 (Unit 12)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Submit empty form	Name: (empty), Model:(empty)	System should show error and not allow submission	Pass
2.	Submit form with partially empty fields	Name: Tesla, Model: (empty)	System shows error for missing field	Pass

Unit Testing 13: Battery Validation

Testing Objective: To ensure battery field accepts only valid values

Table 99:Test Case-M4 (Unit 13)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Enter negative value	Battery: -50	System don't give right to enter negative value for battery	Pass
2.	Enter text	Battery: ABC	System don't give right to enter any other value except number	Pass

Unit Testing 14: Show Saved Vehicles

Testing Objective: To ensure system retrieves saved vehicle records correctly.

Table 100:Test Case-M4 (Unit 14)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Fetch vehicle list from system	User with saved vehicles	System show all stored vehicles	Pass
2.	Fetch vehicle list after adding new vehicle	Newly added vehicle exists	System returns updated list including new vehicle	Pass

3.	Fetch vehicle list with multiple vehicles	3–4 vehicles saved	System returns complete list of all vehicles	Pass
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Unit Testing 15: Empty Vehicle List Handling

Testing Objective: To ensure system handles no-data condition properly

Table 101:Test Case-M4 (Unit 15)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	View vehicles when no data exists	No vehicles in account	System shows “No vehicles Found “ message	Pass
2.	Remove all vehicles and view list	Previously deleted all vehicles	System shows empty state message	Pass
3.	First-time user access	New user account	System shows no vehicle data available	Pass

Unit Testing 16: User-Specific Data Validation

Testing Objective: To ensure only user’s own vehicles are retrieved

Table 102:Test Case-M4 (Unit 16)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Login with User A and fetch vehicles	User a account	Only User A’s vehicles are displayed	Pass
2.	Login with User B and fetch vehicles	User B account	Only User B’s vehicles are displayed	Pass

3.	Switch between users	User A → User B	System updates and shows respective user data	Pass
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Unit Testing 17: Update Vehicle Data

Testing Objective: To ensure vehicle data can be updated correctly

Table 103:Test Case-M4 (Unit 17)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update vehicle model	Model: Model X	System updates vehicle data successfully	Pass
2.	Update multiple fields	Model: Model Y, Battery: 80	System updates all fields correctly	Pass
3.	Update with same data	No changes made	System keeps existing data	Pass

Unit Testing 18: Invalid Update Validation

Testing Objective: To ensure invalid data is not accepted

Table 104:Test Case-M4 (Unit 18)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Enter invalid battery	Battery: -100	System shows error and rejects update	Pass
2.	Enter text in numeric field	Battery: ABC	System shows validation error	Pass
3.	Leave required field empty during update	Name: (empty)	System shows error and prevents update	Pass

Unit Testing 19: Delete Vehicle Record

Testing Objective: To ensure vehicle can be deleted successfully

Table 105:Test Case-M4 (Unit 19)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Delete selected vehicle	Vehicle ID selected	System removes vehicle from system	Pass
2.	Delete one vehicle from multiple records	3 vehicles exist	Only selected vehicle is deleted	Pass
3.	Delete last remaining vehicle	Only 1 vehicle exists	System deletes vehicle and list becomes empty	Pass

Unit Testing 20: Delete Without Selection

Testing Objective: To ensure system handles invalid delete action

Table 106:Test Case-M4 (Unit 20)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Click delete without selecting vehicle	No selection	System shows warning message	Pass
2.	Click delete with invalid ID	Invalid Vehicle ID	System shows error message	Pass
3.	Try delete without permission (other user data)	Unauthorized user	System blocks action	Pass

Unit Testing 21: Select Vehicle Successfully

Testing Objective: To ensure user can successfully select a vehicle for route planning

Table 107:Test Case-M4 (Unit 21)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Select a vehicle for route planning	Vehicle ID selected	System selects the vehicle successfully for route planning	Pass
2.	Select vehicle from multiple vehicles	3 vehicles exist	System allows selection of only one vehicle	Pass
3.	Change selected vehicle	Another vehicle selected	System updates selection and replaces previous vehicle	Pass

Unit Testing 22: Charging Rate Display Function

Testing Objective: To ensure that the system correctly displays charging rates from different stations for comparison.

Table 108: Test Case-M5 (Unit 22)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Display charging rate of Station A	Rate: PKR 85/unit	Correct charging rate displayed	Pass
2.	Display charging rate of Station B	Rate: PKR 95/unit	Correct charging rate displayed	Pass
3.	Compare rates of multiple stations	Multiple station rates available	Charging rates displayed side by side for comparison	Pass

Unit Testing 23: Travel Expense Estimation

Testing Objective: To ensure that the system helps users estimate travel expenses before beginning the journey.

Table 109: Test Case-M5 (Unit 23)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Use station price for expense estimation	Station Rate: PKR 85/unit	Relevant cost information available for estimation	Pass
2.	Display route suggestions using GPS/map integration	Distance: 5 km / 9 km	Distance information shown correctly	Pass

Unit Testing 24: Station Registration with Valid Data

Testing Objective: To ensure that a charging station is registered successfully with valid input

Table 110:Test Case-M6 (Unit 24)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Register station with all valid fields	Station Name: Green Charge, Type: Fast Charger, Location: valid	Station registered successfully and pending approval	Pass
2.	Register station with missing required field	Station Name: (empty), Type: Fast Charger	System shows error and does not allow submission	Pass
3.	Register station with duplicate station name	Station Name: already exists in system	System shows duplicate entry error	Pass
4.	Register station without owner login	No authenticated session	System blocks access and redirects to login	Pass

Unit Testing 25: Station Submission for Approval

Testing Objective: To ensure station details are validated before submission for approval

Table 111:Test Case-M6 (Unit 25)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Submit station with all required details provided	All fields filled; documents uploaded	Station submitted for admin approval	Pass
2.	Submit station without providing station details	Station details: (empty)	System shows validation error	Pass
3.	Submit station and verify dependency on FR-2 (login)	User not logged in	System denies access and shows login prompt	Pass

Unit Testing 26: Update Charger Type

Testing Objective: To ensure station owner can update charger type details successfully

Table 112:Test Case-M6 (Unit 26)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update charger type to a new valid type	Charger Type: DC Fast Charger	System updates charger type and saves changes	Pass
2.	Update charger type with empty value	Charger Type: (empty)	System shows validation error	Pass
3.	Other user attempts to update station details	Different owner account	System blocks action with unauthorized error	Pass
4.	Update station details for unregistered station	Station ID: invalid/non-existent	System shows station not found error	Pass

Unit Testing 27: Owner Authorization Validation

Testing Objective: To ensure only the station owner can update their own station details

Table 113:Test Case-M6 (Unit 27)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Login as owner and update own station	Owner: Ali, Station: Green Charge	System allows update successfully	Pass
2.	Login as different owner and attempt update	Owner: Sara, Station: Green Charge (owned by Ali)	System shows unauthorized access error	Pass
3.	Update station without authentication	No valid session token	System denies request and shows error	Pass

Unit Testing 28: Valid Price Update

Testing Objective: To ensure station owner can update charging price with valid values

Table 114:Test Case-M6 (Unit 28)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update price with valid numeric value	Price: 15 PKR/kWh	Price updated and applied immediately	Pass
2.	Update price with zero value	Price: 0	System shows error - price must be greater than zero	Pass
3.	Update price with negative value	Price: -5	System shows validation error	Pass
4.	Update price with alphabetic characters	Price: ABC	System rejects input and shows error	Pass

Unit Testing 29: Immediate Price Application

Testing Objective: To ensure updated prices are applied immediately after saving

Table 115:Test Case-M6 (Unit 29)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update price and verify it reflects for EV users	Price changed from 10 to 15 PKR/kWh	New price immediately visible to EV users	Pass
2.	Update price and check previous sessions are unaffected	Ongoing charging session exists	Old rate applies to ongoing session, new rate for new ones	Pass

Unit Testing 30: Availability Status Update

Testing Objective: To ensure station owner can update availability status in real time

Table 116:Test Case-M6 (Unit 30)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Set station status to Available	Status: Available	System updates status in real time	Pass
2.	Set station status to Unavailable/Maintenance	Status: Under Maintenance	System updates status and reflects to EV users instantly	Pass
3.	Submit availability update without selecting a status	Status: (empty/not selected)	System shows validation error	Pass
4.	Verify real-time update visible to EV users	EV user views station list after status change	Updated availability status shown correctly	Pass

Unit Testing 31: Unauthorized Availability Update

Testing Objective: To ensure unauthorized users cannot change station availability

Table 117:Test Case-M6 (Unit 31)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Non-owner user attempts availability update	Different user account	User must have registered or login to use app	Pass
2.	Update availability without login	Register or login	First user has to login then other details are displayed	Pass
3.	Update availability for non-existent station	Station ID: invalid	System shows station not found error	Pass

Unit Testing 32: Valid Operating Hours Update

Testing Objective: To ensure station owner can update operating hours with valid 24-hour format

Table 118:Test Case-M6 (Unit 32)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update hours with valid 24-hour format	Open: 08:00, Close: 20:00	System saves operating hours successfully	Pass
2.	Enter overlapping or invalid time range	Open: 22:00, Close: 06:00 (next day invalid)	System shows error for invalid time range	Pass
3.	Enter operating hours in 12-hour format	Open: 8 AM, Close: 8 PM	System rejects input and requires 24-hour format	Pass
4.	Leave operating hours fields empty	Open: (empty), Close: (empty)	System shows required field validation error	Pass

Unit Testing 35: Profile Information Retrieval

Testing Objective: To ensure authenticated EV owners can view their personal profile information

Table 119:Test Case-M7(Unit 35)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	View profile as authenticated EV owner	User: Sara Ahmed, logged in	System displays Sara's profile details correctly	Pass
2.	Attempt to view profile without login	No valid session	User don't have access to view profile without login	Pass
4.	View profile and verify all fields are displayed	Complete profile with name, email, phone	All stored profile fields are shown correctly	Pass

Unit Testing 36: Profile Data Accuracy Validation

Testing Objective: To ensure profile information displayed is accurate and up to date

Table 120:Test Case-M7 (Unit 36)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	View profile after updating information	Name updated to new value	Updated name reflected in profile view	Pass
2.	View profile with incomplete data	Phone number: (empty)	System displays available fields; empty fields shown as blank	Pass
3.	View profile on multiple devices simultaneously	Same account, two sessions	Consistent profile data displayed on both	Pass

Unit Testing 37: Valid Profile Update

Testing Objective: To ensure authenticated EV owners can update their own profile information

Table 121:Test Case-M7 (Unit 37)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update profile with valid new name	Name: Sara Ali	Profile updated and saved successfully	Pass
2.	Update profile with empty required field	Name: (empty)	System shows validation error and blocks update	Pass
3.	Attempt to update another user's profile	User A edits User B profile data	System don't give access to another user's account	Pass
4.	Update email with invalid format	Email: sara@invalid	System shows invalid email format error	Pass

Unit Testing 38: Profile Update Authorization

Testing Objective: To ensure only the owner can update their own profile information

Table 122:Test Case-M7 (Unit 38)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Update own profile while logged in	Valid session, own profile	System allows and saves the update	Pass
2.	Update profile without authentication	No session token	System denies request and redirects to login	Pass
3.	Update profile and verify changes persist after logout and re-login	Updated name: Ayesha Khan	Updated info retained after fresh login	Pass

Unit Testing 39: System Performance Monitoring

Testing Objective: To ensure the administrator can view accurate system performance information and handle unavailable performance data correctly.

Table 123:Test Case-M8 (Unit 39)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	View system performance dashboard	Last 24 Hours	System displays CPU usage, memory usage, active users and server status correctly	Pass

2.	View performance data for unavailable period	No performance records	System displays "No performance data available" message	Pass
3.	Refresh system performance dashboard	Refresh Button	Dashboard reloads and displays the latest performance information	Pass

Unit Testing 40: Station Approval Management

Testing Objective: To ensure administrators can approve valid charging stations and reject invalid station requests.

Table 124:Test Case-M8 (Unit 40)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Approve pending charging station	Station ID: CS-1003	CS-1003 Station status changes to Approved and becomes visible to users	Pass
2.	Reject invalid charging station	Station ID: CS-1004 (Incomplete Data)	Station request is rejected and remains unavailable	Pass
3.	Approve an already approved station	Station ID: CS-1003	System prevents duplicate approval and displays a warning message	Pass

Unit Testing 41: Rating and Review Input Validation and Sentiment Processing

Testing Objective: To ensure that rating, written review, sentiment analysis, spam detection, and review storage functions work correctly.

Table 125:Test Case-M9 (Unit 41)

N o.	Test case/Test script	Attribute and value	Expected result	Result
1	Validate rating value within allowed range	Rating: 1 to 5	Rating accepted as valid	Pass
2	Reject rating value outside allowed range	Rating: 0 or 6	Validation error displayed	Pass

3	Validate written review text	Review: meaningful sentence	Review text accepted for analysis	Pass
4	Detect empty review text	Review: blank	Error message displayed	Pass
5	Run ML sentiment classification	Review text: positive/neutral/negative	Sentiment class generated correctly	Pass

Unit Testing 42: Review Moderation and Rating Calculation

Testing Objective: To ensure that spam/fake review flagging, overall rating calculation, and admin moderation functions work correctly.

Table 126: Test Case-M9 (Unit 42)

No.	Test case/Test script	Attribute and value	Expected result	Result
1	Detect spam or toxic review	Repeated/offensive review text	Review flagged or rejected	Pass
2	Compare star rating with ML sentiment	Rating: 5, sentiment: negative	Inconsistent review flagged	Pass
3	Calculate overall station rating	Ratings: 4, 5, 3	Correct average rating calculated	Pass
4	Store accepted review with metadata	User ID, station ID, rating, review, sentiment	Review saved with correct station and user reference	Pass
5	Allow admin review removal	Admin role, flagged review ID	Review removed successfully	Pass

Unit Testing 43: Amenities Display, Update, Permission, and Filtering

Testing Objective: To ensure that amenities fetching, updating, permission checking, and filtering functions work correctly.

Table 127:Test Case-M10 (Unit 43)

No.	Test case/Test script	Attribute and value	Expected result	Result
1	Fetch amenities by station ID	Station ID: ST-101	Correct amenities list displayed	Pass
2	Handle station with no amenities	Station ID with empty amenities list	No amenities message displayed	Pass
3	Validate owner permission for amenities update	Owner ID matches station owner ID	Update permission granted	Pass
4	Reject unauthorized amenities update	Different owner/user account	Permission error displayed	Pass
5	Filter stations using selected amenities	Filter: Wi-Fi + Parking	Only matching stations returned	Pass

2.14.2 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the module is tested. This is to ensure that the system produced meets the specifications and requirements.

Functional Testing 1: User Registration, Login, and Password Recovery

Objective: To ensure that users can register with email, login securely, recover passwords, and access the system only after successful authentication.

Table 128:Test Case-M1 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Register a new user account	Email: newuser@gmail.com Password: newuser@12	User account created successfully	Pass
2.	Login with valid registered credentials	Email: newuser@gmail.com Password: newuser@12	User logged in successfully and authorized area loaded	Pass
3.	Login with invalid credentials	Valid email + wrong password	Error message “Invalid credentials” displayed	Pass
4.	Recover password using registered email	Email: newuser@gmail.com	Password recovery option works successfully	Pass
5.	Access system without authentication	No login	System restricts access and redirects to login page	Pass

Functional Testing 2: Charging Station Locator

Objective: To ensure that the system correctly finds nearby charging stations, shows their details, and applies filters according to user needs.

Table 129:Test Case-M2 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Display nearby charging stations	GPS enabled	Nearby stations displayed on map	Pass
2.	Show charging station details	Valid station selected	Station name, charger type, cost per unit, and status displayed	Pass
3.	Filter stations by location	Location selected	Only stations from selected location displayed	Pass
4.	Filter stations by charger compatibility	Charger type selected	Only compatible stations displayed	Pass
5.	Filter stations by price	Price range selected	Stations within selected price range displayed	Pass

Functional Testing 3: Route Planning

Objective: To ensure that the system calculates remaining travel distance using battery percentage and selected vehicle details, then suggests reachable charging stations or shows an alert when none are available.

Table 130:Test Case-M3 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Enter current battery percentage	Battery: 35%	Battery percentage accepted successfully	Pass
2.	Use selected vehicle details	Selected vehicle available	Vehicle model, battery capacity, and range loaded successfully	Pass
3.	Calculate remaining travel distance	Battery % + vehicle range	Travel distance calculated correctly	Pass
4.	Suggest nearest charging stations	GPS enabled	Nearest reachable charging stations displayed	Pass
5.	Generate alert when no station is available	No station in remaining range	User alerted to fully charge vehicle at home before traveling	Pass

Functional Testing 4: Vehicle Profile Management (Add, View, Edit, Delete)

Objective: To ensure that users can successfully manage multiple vehicle profiles including adding, viewing, updating, and deleting vehicle information

Table 131:Test Case-M4 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Add a new vehicle profile	Name: Toyota Corolla, Model: 2023, Battery: 50 kWh	Vehicle profile is added successfully	Pass
2.	View vehicle profiles	Existing vehicles in account	All saved vehicle profiles are displayed correctly	Pass
3.	Edit vehicle profile	Updated Model:2023 to 2024	Vehicle information is updated successfully	Pass
4.	Delete vehicle profile	Selected Vehicle ID	Vehicle profile removed successfully	Pass
5.	Add multiple vehicle profiles	2–3 vehicles added	System allows multiple vehicles to be stored	Pass
6.	Store vehicle details correctly	Name, Model, Battery Capacity	All vehicle data is saved and displayed correctly	Pass

Functional Testing 5: Cost Comparison

Objective: To ensure that users can compare charging rates from different stations and use that information to estimate travel expenses before starting the journey.

Table 132:Test Case-M5 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Display charging rates from different stations	Multiple station rates available	Charging rates displayed successfully	Pass
2.	Compare stations using price information	Different station prices	User can compare charging rates correctly	Pass
3.	Identify cheapest option	Cost data	Cheapest station highlighted	Pass
4.	Compare with invalid data	Missing values	Error displayed	Pass

Functional Testing 6: Station Registration and Manage Station details

Objective: To ensure station owners can create accounts, manage charging station details.

Table 133:Test Case-M6 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Register station owner and charging station	Owner Name, Station Name, Charger Type	Owner account and station are created successfully	Pass
2.	Update station charging details	Opening: 08:00, Closing: 20:00	Station details are updated successfully	Pass
3.	Change station availability	Updated Model:2023 to 2024	Operating hours are updated correctly	Pass
4.	Delete vehicle profile	Available, Unavailable	Station status is updated successfully	Pass
5.	View updated station information	Station profile data	Updated station details are displayed correctly	Pass

Functional Testing 7: Profile & Vehicle Access Management

Objective: To ensure EV owners can manage profiles, access vehicle data, and provide feedback successfully.

Table 134:Test Case-M7 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Manage user profile	Name, Email, Password	Profile is updated successfully	Pass
2.	View personal profile	Logged-in EV owner	All saved vehicle profiles are displayed correctly	Pass
3.	Access vehicle profiles	Multiple saved vehicles	Profile information is displayed correctly	Pass
4.	Select vehicle for use	Vehicle ID selected	Vehicle is selected successfully	Pass
5.	Submit rating and review	Rating: 4/5, Review text	Feedback is submitted successfully	Pass
6.	Update profile information	Name, Email updated	Profile changes are saved successfully	Pass

Functional Testing 8: Admin Management Portal

Objective: To ensure administrators can manage charging stations, user accounts, system performance, and station approval functions successfully.

Table 135:Test Case-M8 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
7.	Add new charging station	Station Name, Location, Charger Type, Cost	Charging station is added successfully and appears in the station list	Pass
8.	Update charging station information	Updated Cost, Availability, Charger Type	Station information is updated successfully	Pass
9.	Delete charging station	Station ID: CS-1003	Selected charging station was removed successfully Profile information is displayed correctly	Pass

10. System performance statistics are displayed correctly	View system performance dashboard	Last 24 Hours Performance Data	Vehicle is selected successfully	Pass
11.	Create user account	Username, Email, User Role	New user account is created successfully	Pass
12.	Approve newly added station	Station ID: CS-1004 (Pending)	Station is approved successfully and becomes available to users	Pass

Functional Testing 9: Rating Submission, Review Display, ML Filtering, and Admin Moderation

Objective: To ensure that EV Owners can submit ratings/reviews, the ML model analyzes feedback, and Admin can remove inappropriate reviews.

Table 136: Test Case-M9 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
1	Submit a valid rating for a station	Rating: 5 stars	Rating saved successfully	Pass
2	Submit written review for a station	Review: "Fast charger and clean station"	Review submitted for ML analysis	Pass
3	Display overall rating and reviews	Station has 5 approved reviews	Average rating and comments displayed	Pass
4	Analyze review sentiment using ML	Positive review text	Sentiment stored as Positive	Pass
5	Flag spam or inappropriate review	Offensive/repeated review text	Review flagged or rejected	Pass
6	Admin removes inappropriate review	Flagged review selected by Admin	Review removed from public list	Pass

Functional Testing 10: Amenities Display, Owner Update, and Amenity-Based Filtering

Objective: To ensure that amenities are displayed correctly, station owners can update amenities, and EV Owners can filter stations using required amenities.

Table 137: Test Case-M10 (Functional)

No.	Test case/Test script	Attribute and value	Expected result	Result
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1	Display station amenities	Amenities: Wi-Fi, Restroom, Café	Amenities visible on station detail page	Pass
2	Update amenities as station owner	Add Parking and Rest Area	Amenities updated successfully	Pass
4	Filter stations by amenities	Filter: Restroom + Parking	Only matching stations displayed	Pass
5	Apply filter with no matching station	Filter combination with no station	No-results message displayed	Pass

2.14.3 Integration Testing

Table 138: Test Case-1 (Integration)

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Login as EV Owner and access charging station locator	Email: evuser@gmail.com Password: 123456	Login successful and EV Owner dashboard is loaded. User is able to access charging station locator and nearby stations are displayed on map.	Pass
2.	Login and plan route using vehicle data	Email: evuser@gmail.com Password: 123456 Vehicle: Selected Battery: 40%	User logged in successfully. Route planning module retrieves vehicle data and calculates remaining distance correctly.	Pass
3.	Plan route with low battery and select stations from locator	Battery: 20% Destination: Lahore	Route planning module calculates route and fetches nearby charging stations from station locator. Route displayed with charging stops.	Pass
4.	Select charging stations and compare cost	Selected Stations: Station A, Station B	System retrieves station data (cost per unit, type) from locator and displays cost comparison correctly.	Pass
5.	Plan route and estimate charging cost	Destination: Lahore Battery: 30%	System generates route and uses station data to estimate total charging cost for the trip.	Pass
6.	Access route planning after login	Email: evuser@gmail.com Password: 123456	Login successful and user is able to access	Pass

			route planning module and perform operations.	
7.	Access cost comparison using station data	Selected stations from locator	System retrieves selected station data and displays cost comparison results successfully.	Pass
8.	Attempt to access station locator without login	No login session	System restricts access and redirects user to login page.	Pass
9.	Route planning when no stations available in locator	Remote location Battery: 15%	System fails to find stations from locator and displays alert advising user that no station available.	Pass
10.	Station owner updates station data and user views updated data	Updated cost/status by station owner	Updated station data is reflected in locator and cost comparison modules for EV Owner.	Pass
11.	Submit review and update station details view	EV Owner submits rating/review for Station A	Review stored, ML result generated, and station detail page shows updated rating/review	Pass
12.	Review module communicates with ML API/service	Review submitted	ML service returns sentiment and spam/inappropriate flag to review module	Pass
13.	Admin moderation updates EV Owner review display	Admin removes flagged review	Removed review no longer appears in station detail page and rating summary updates	Pass
14.	Station owner updates amenities and EV Owner views station	Owner adds Restroom and Café	Updated amenities appear correctly on EV Owner station detail page	Pass
15.	Amenities filter integrates with station locator	EV Owner applies Parking filter	Station locator returns only stations with Parking amenity	Pass

2.14.4 Final Testing Summary Paragraph

The Smart EV Charging and Navigation Management System was tested using Appium-based mobile automation testing. The testing focused on the main functional modules of the application, including role selection, authentication, nearby charging station search, map and location permission, search and filter options, station details, vehicle management, route planning, charging cost estimation, reviews and ratings, user profile, and logout. Both positive and negative test cases were prepared to verify that the

application works correctly with valid inputs and also handles invalid inputs, empty fields, permission denial, unavailable stations, low battery, and network issues properly. The main purpose of Appium testing was to validate the application from the end user's point of view and confirm that all important mobile app flows work smoothly without crashes or incorrect navigation.